

DERIVATIVES PRICING

A COMPUTATIONAL & THEORETICAL FRAMEWORK

©:Zac Kienzle[†]

[†] University of Queensland, Brisbane, Australia

Contents

I	Mathematical Foundations	6
1	Measure and Integration	6
2	Probability	6
2.1	Modelling Random Experiments	6
2.2	Random Variables	8
2.3	Probability Distributions	9
2.4	Expectation	12
2.5	L^p Spaces	14
2.6	Integral Transforms	15
2.7	Information and Independence	17
2.8	Important Stochastic Processes	19
3	Convergence	22
4	Conditioning	22
5	Martingales	22
6	Wiener and Brownian Motion Processes	22
7	Itô Calculus	22
7.1	Itô Integral	22
7.1.1	Itô Integral for Predictable Processes	25
7.2	Itô Calculus	30
7.3	Stochastic Differential Equations	32
7.3.1	Methods for Solving Simple SDEs	34
7.3.1.1	Linear Stochastic Differential Equations	34

II	Levy Processes	36
III	Stochastic Volatility	37
IV	Risk-Neutral Valuation	37
8	Discrete-time Market Model	37
9	Continuous-time Market Model	37
V	Martingale Methods	37
10	The Black-Scholes Model	37
11	A Dividend-paying Stock	37
12	Discretisation Schemes	37
13	Connection to Partial Differential Equations	38
VI	Monte Carlo Methods	38
14	Generating Random Numbers and Random Variables	39
15	Quasi Monte Carlo	39
15.1	Discrepancy Measures	39
15.2	Low-discrepancy Sequences	39
15.2.1	Halton and Hammersley	39
15.2.2	Faure	39
15.2.3	Sobol	39
15.3	Koksma-Hlawka	39
16	Variance Reduction	39

VII	Grid & Lattice Methods	39
17	Lattice Methods	39
18	Finite Difference Methods	40
19	Contract Sensitivities	40
19.1	Likelihood Ratio	40
19.2	Pathwise Derivatives	40
19.3	Finite-Difference Approximations	40
VIII	Financial Derivatives	40
20	European Options	40
21	Exotic Options	40
21.1	Introduction	40
21.2	Maximum of Brownian Motion with Drift	41
21.3	Barrier Options	45
21.4	Knock-out Barrier	45
21.4.1	Up-and-out barrier call option	45
21.4.2	Up-and-out barrier put option	48
21.4.3	Down-and-out barrier call option	50
21.4.4	Down-and-out barrier put option	52
21.5	Knock-in Barrier	54
21.5.1	Up-and-Out Call	54
21.6	Lookback Options	54
21.7	The Lookback Put Option	54
21.8	Floating Strike Lookback Option	55
21.8.1	Floating Strike Lookback Option	55
21.9	Asian Options	55

22 American Options	58
23 Basket Options	58
A Existence of Lebesgue Measure	59
B Linear Algebra and Functional Analysis	59
C Multivariate Differentiation	59
D Probability and Statistics	59
E Python Primer	59

Part I

Mathematical Foundations

1 Measure and Integration

2 Probability

2.1 Modelling Random Experiments

Probability theory is about modelling and analysing *random experiments*: experiments whose outcome cannot be determined in advance, but are nevertheless still subject to analysis. Mathematically, we can model a random experiment by defining a specific measure space $(\Omega, \mathcal{H}, \mathbb{P})$, which we call a *probability space*, where the three components are: a *sample space*, a collection of *events*, and a *probability measure*.

The *sample space* Ω of a random experiment is the set of all possible outcomes of the random experiment. Sample spaces can be countable, such as $\mathbb{N}$, or uncountable, such as $\mathbb{R}^n$ or $\{0, 1\}^\infty$. The sets in the σ -algebra $\mathcal{H}$ are called *events*. These are the subsets to which we wish to assign a probability. We can view $\mathcal{H}$ as the family of events. An event A *occurs* if the outcome of the experiment is one of the elements in A . The properties of $\mathcal{H}$ are natural in the context of random experiments and events:

- If A and B are events, the set $A \cup B$ is also an event, namely the event that A *or* B or both occur. For a sequence $A_1, A_2, \dots$ of events, their union is the event that *at least one* of the events occurs.
- If A and B are events, the set $A \cap B$ is also an event, namely the event that A *and* B both occur. For a sequence $A_1, A_2, \dots$ of events, their intersection is the event that *all* of the events occur.
- If A is an event, its complement A^c is also an event, namely the event that A *does not* occur.

- The set Ω itself is an event, namely the *certain event* (it always occurs). Similarly, $\emptyset$ is an event, namely the *impossible event* (it never occurs).

When the sample space Ω is $\mathbb{R}$, the natural σ -algebra $\mathcal{H}$ is the *Borel σ -algebra* $\mathcal{B}$. Recall that this is the smallest σ -algebra on $\mathbb{R}$ that contains all the intervals of the form $(-\infty, x]$ for $x \in \mathbb{R}$. This σ -algebra of sets is big enough to contain all important sets and small enough to allow us to assign a probability to all events.

The third ingredient in the model for a random experiment is the specification of the probability of the events. In fact, this is the crucial part of the model. Mathematically, we are looking for a *measure* $\mathbb{P}$ that assigns to each event A a number $\mathbb{P}(A)$ in $[0, 1]$ describing how likely or probable that event is.

Definition 2.1: Probability Measure

A *probability measure* is a measure on $(\Omega, \mathcal{H})$ with $\mathbb{P}(\Omega) = 1$. Thus, $\mathbb{P}$ is a mapping from $\mathcal{H}$ to $[0, 1]$ with the following properties:

- (1) $\mathbb{P}(\emptyset) = 0$ and $\mathbb{P}(\Omega) = 1$.
- (2) For any sequence $A_1, A_2, \dots$ of disjoint events,

$$\mathbb{P}(\cup_n A_n) = \sum_n \mathbb{P}(A_n). \quad (2.1)$$

An event A is said to occur *almost surely* (a.s.) if $\mathbb{P}(A) = 1$.

Theorem 2.1: Properties of a Probability Measure

Let $\mathbb{P}$ be a probability measure on a measurable space $(\Omega, \mathcal{H})$, and let A, B and $A_1, A_2, \dots$ be events. Then, the following hold:

- (1) (Monotonicity): $A \subseteq B \implies \mathbb{P}(A) \leq \mathbb{P}(B)$.
- (2) (Continuity from below): If $A_1, A_2, \dots$ is an increasing sequence of events, i.e., $A_1 \subseteq A_2 \subseteq \dots$, then

$$\lim \mathbb{P}(A_n) = \mathbb{P}(\cup_n A_n). \quad (2.2)$$

- (3) (Countable subadditivity): $\mathbb{P}(\cup_n A_n) \leq \sum_n \mathbb{P}(A_n)$.

Theorem 2.2: Additional Properties of a Probability Measure

Let $\mathbb{P}$ be a probability measure on a measurable space $(\Omega, \mathcal{H})$, and let A, B and $A_1, A_2, \dots$ be events. Then, the following hold:

- (1) (Complement): $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
- (2) (Union): $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.
- (3) (Continuity from above): If $A_1, A_2, \dots$ is a decreasing sequence of events, i.e., $A_1 \supseteq A_2 \supseteq \dots$, then

$$\lim \mathbb{P}(A_n) = \mathbb{P}(\cap_n A_n). \quad (2.3)$$

2.2 Random Variables

Usually, the most convenient way to describe quantities of interest connected with random experiments is by using random variables. Random variables allow us to use intuitive notations for certain events, such as $\{X \in A\}$, etc. Intuitively, we can think of a random variable X as a measurement on a random experiment that will become available *tomorrow*. However, all the think work can be done *today*. If for each possible outcome ω , the measurement is $X(\omega)$, we can specify today probabilities such as $\mathbb{P}(X \in A)$.

We can translate our intuitive notion of a random variable into rigorous mathematics by defining a random variable to be a measurable function from $(\Omega, \mathcal{H})$ to a measurable space $(E, \mathcal{E})$.

Definition 2.2: Random Variable

Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space and $(E, \mathcal{E})$ a measurable space. A *random variable* X taking values in E is an $\mathcal{H}/\mathcal{E}$ -measurable function; that is, a mapping $X : \Omega \rightarrow E$ that satisfies

$$X^{-1}A = \{X \in A\} = \{\omega \in \Omega : X(\omega) \in A\} \in \mathcal{H} \quad \text{for all } A \in \mathcal{E}. \quad (2.4)$$

Often a random experiment is described via more than one random variable. Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space and let $\mathbb{T}$ be an arbitrary index set, which may be countable or uncountable. Think of $\mathbb{T}$ as a set of times. For each $t \in \mathbb{T}$, let $(E_t, \mathcal{E}_t)$ be a measurable

space. The product measurable space of $\{(E_t, \mathcal{E}_t) : t \in \mathbb{T}\}$ is

$$\left(\times_{t \in \mathbb{T}} E_t, \bigotimes_{t \in \mathbb{T}} \mathcal{E}_t \right).$$

Each element $x \in \times_{t \in \mathbb{T}} E_t$ is of the form $x = (x_t)_{t \in \mathbb{T}}$. Often $(E_t, \mathcal{E}_t)$ is the same space $(E, \mathcal{E})$ for all t , in which case x can be viewed as a function $x : \mathbb{T} \rightarrow E$. The σ -algebra on $\times_{t \in \mathbb{T}} E_t$ is generated by the rectangles

$$R((A_t)_{t \in \mathbb{T}}) := \left\{ x \in \times_{t \in \mathbb{T}} E_t : x_t \in A_t \text{ for each } t \in \mathbb{T} \right\}, \quad (2.5)$$

where $A_t \in \mathcal{E}_t$ and $A_t = E_t$ except for finitely many t .

Definition 2.3: Stochastic Process

A collection of random variables $X := \{X_t, t \in \mathbb{T}\}$, where $\mathbb{T}$ is any index set, is called a *stochastic process*. When $\mathbb{T}$ is finite, X is called a *random vector*.

Theorem 2.3: Stochastic Process

$X := \{X_t, t \in \mathbb{T}\}$ is a stochastic process taking values in $\times_{t \in \mathbb{T}} E_t$ with σ -algebra $\bigotimes_{t \in \mathbb{T}} \mathcal{E}_t$ if and only if each X_t is a random variable taking values in E_t with σ -algebra $\mathcal{E}_t$.

2.3 Probability Distributions

Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space and let X be a random variable taking values in a set E with σ -algebra $\mathcal{E}$. To simplify our usage, we also say that “ X takes values in $(E, \mathcal{E})$ ”. If we wish to describe our experiment via X , then we need to specify the probabilities of events such as $\{X \in B\}$, $B \in \mathcal{E}$. In elementary probability theory we usually made a distinction between “discrete” and “continuous” random variables. However, it is sometimes better to analyse random variables in a more unified manner; many definitions and properties of random variables do not depend on whether they are “discrete” or “continuous”. Moreover, some random variables are neither discrete nor continuous.

Recall that by definition *all* the sets $\{X \in B\}$ with $B \in \mathcal{E}$ are events. So we may assign a probability to each of these events. This leads to the notion of the *distribution* of a random variable.

Definition 2.4: Probability Distribution

Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space and let X be a random variable taking values in $(E, \mathcal{E})$. The (*probability*) *distribution* of X is the image measure μ of $\mathbb{P}$ under X . That is,

$$\mu(B) := \mathbb{P}(X \in B), \quad B \in \mathcal{E}. \quad (2.6)$$

Definition 2.5: Cumulative Distribution Function

The (*cumulative*) *distribution function* of a numerical random variable X is the function $F : \mathbb{R} \rightarrow [0, 1]$ defined by

$$F(x) := \mathbb{P}(X \leq x), \quad x \in \mathbb{R}. \quad (2.7)$$

Proposition 2.1: Properties of a Cumulative Distribution Function

- (1) (*Bounded*): $0 \leq F(x) \leq 1$.
- (2) (*Increasing*): $x \leq y \implies F(x) \leq F(y)$.
- (3) (*Right-continuous*): $\lim_{h \downarrow 0} F(x + h) = F(x)$.

Definition 2.6: Discrete Distribution

We say that X has a *discrete distribution* μ on $(E, \mathcal{E})$ if μ is a discrete measure; that is, it is of the form

$$\mu = \sum_{x \in D} f(x) \delta_x$$

for some countable set D and positive masses $\{f(x), x \in D\}$.

Definition 2.7: Distribution with a Density

We say that X has an *absolutely continuous* distribution μ on $(E, \mathcal{E})$ with respect to a measure λ on $(E, \mathcal{E})$ if there exists a positive $\mathcal{E}$ -measurable function f such that $\mu = f\lambda$; that is,

$$\mu(B) = \mathbb{P}(X \in B) = \int_B \lambda(dx) f(x), \quad B \in \mathcal{E}.$$

The function f is thus the (*probability*) *density function* (pdf) of X with respect to the measure, λ often taken to be the Lebesgue measure. Note that μ is absolutely continuous with respect to λ in the sense that for all $B \in \mathcal{E} : \lambda(B) = 0 \implies \mu(B) = 0$.

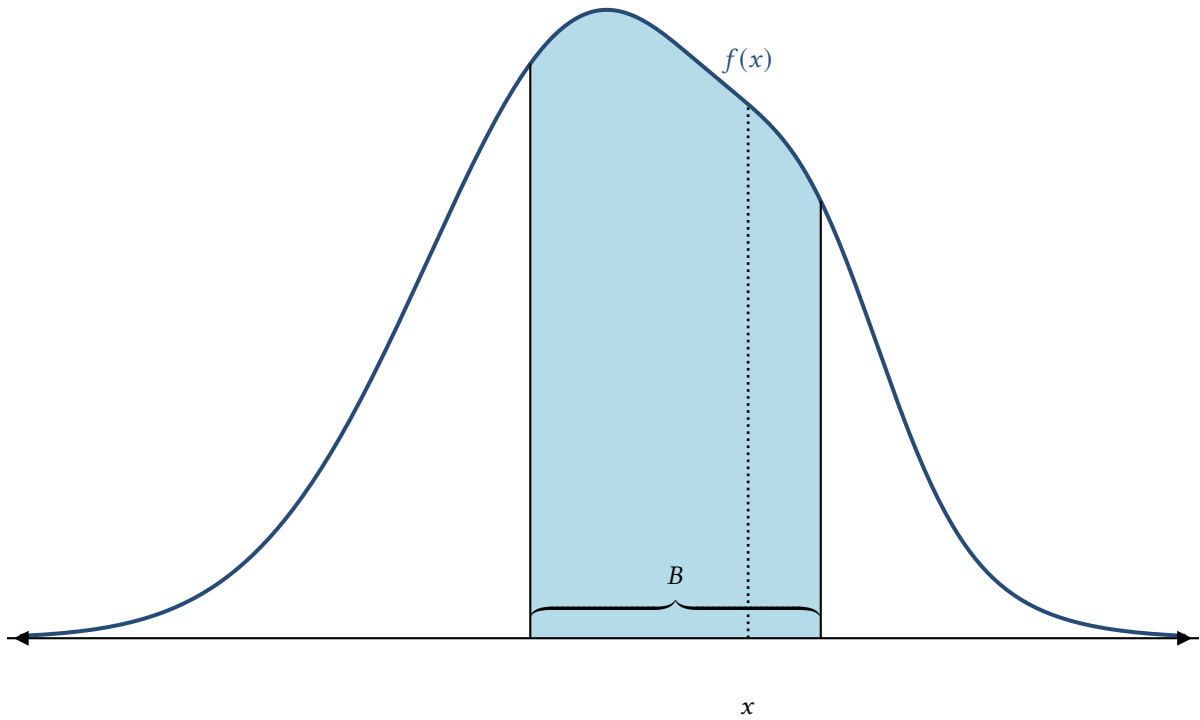


Figure 1: A probability density function of a numerical random variable

Describing an experiment via a random variable and its probability distribution or density is often more convenient than specifying the probability space. In fact, the probability space usually stays in the background. The question remains, however, whether there *exists* a probability space $\Omega, \mathcal{H}, \mathbb{P}$ and a numerical random variable X for a given probability distribution. Fortunately, existence can be established for all practical probability models.

The concept of independence is of great importance in the study of random experiments and the construction of probability distributions. Loosely speaking, independence is about the lack of shared information between random objects.

Definition 2.8: Independent Random Variables

Let X and Y be random variables taking values in $(E, \mathcal{E})$ and $(F, \mathcal{F})$, respectively. They are said to be *independent* if for any $A \in \mathcal{E}$ and $B \in \mathcal{F}$ it holds that

$$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B). \quad (2.8)$$

Intuitively, this means that information regarding X does not affect our knowledge of Y and vice versa. Mathematically, Equation (2.8) simply states that the distribution, μ say, of (X, Y) is given by the *product measure* of the distributions μ_X of X and μ_Y of Y . We can extend the independence concept to a finite or infinite collection of random variables as follows:

Definition 2.9: Independency

We say that a collection of random variables $\{X_t, t \in \mathbb{T}\}$, with each X_t taking values in $(E_t, \mathcal{E}_t)$, is an *independency* if for any finite choice of indexes $t_1, \dots, t_n \in \mathbb{T}$ and sets $A_{t_1} \in \mathcal{E}_{t_1}, \dots, A_{t_n} \in \mathcal{E}_{t_n}$ it holds that

$$\mathbb{P}(X_{t_1} \in A_{t_1}, \dots, X_{t_n} \in A_{t_n}) = \mathbb{P}(X_{t_1} \in A_{t_1}) \cdots \mathbb{P}(X_{t_n} \in A_{t_n}).$$

Attention 2.1: Independence as a Modelling Assumption

More often than not, the independence of random variables is a model *assumption*, rather than a consequence. Instead of describing a random experiment via an explicit description of Ω and $\mathbb{P}$, we will usually model the experiment through one or more (independent) random variables

2.4 Expectation

The expectation of a (numerical) random variable is a weighted average of all the values that the random variable can take. The expectation of a *discrete* random variable X is defined

as:

$$\mathbb{E}X := \sum_x x\mathbb{P}(X = x),$$

and for a *continuous* random variable X with pdf f it is defined as:

$$\mathbb{E}X := \int_{-\infty}^{\infty} xf(x)dx.$$

Having to define the expectation in two ways is unsatisfactory. Also, there are cases wherein random variables are neither discrete nor continuous. How should we define the expectation for these random variables? Here is the formal definition:w4

Definition 2.10: Expectation

Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space and X a numerical random variable. The *expectation* of X is defined as the integral

$$\mathbb{E}X := \mathbb{P}X = \int Xd\mathbb{P}.$$

Theorem 2.4: Properties of an Expectation

Let $a, b \in \mathbb{R}$ and let X, Y , and (X_n) be numerical random variables for which the expectations below are well-defined. Then, the following hold:

- (1) (Monotonicity): If $X \leq Y$, then $\mathbb{E}X \leq \mathbb{E}Y$.
- (2) (Linearity): $\mathbb{E}(aX + bY) = a\mathbb{E}X + b\mathbb{E}Y$.
- (3) (Monotone convergence): If $X_n \uparrow X$, then $\mathbb{E}X_n \uparrow \mathbb{E}X$.

Lemma 2.1: Fatou

For any sequence (X_n) of positive numerical random variables,

$$\mathbb{E} \liminf X_n \leq \liminf \mathbb{E}X_n.$$

Theorem 2.5: Dominated and Bounded Convergence

Let (X_n) be a sequence of numerical random variables for which $\lim X_n = X$ exists.

Then, the following hold:

- (1) (Dominated convergence): If, for each n , $|X_n| \leq Y$ for some random variable Y with $\mathbb{E}Y < \infty$, then $\mathbb{E}X_n \rightarrow \mathbb{E}X < \infty$.
- (2) (Bounded convergence): If $|X_n| \leq c$ for all n for some $c \in \mathbb{R}$, then $\mathbb{E}X_n \rightarrow \mathbb{E}X < \infty$.

Theorem 2.6: Expectation and Image Measure

Let X be a random variable taking values in $(E, \mathcal{E})$ with distribution μ and let h be an $\mathcal{E}$ -measurable function. Then, provided the integral exists:

$$\mathbb{E}h(X) = \int h(X) d\mathbb{P} = \int_{\mathbb{R}} \mu(dx) h(x) = \mu h.$$

Theorem 2.7: Transformation Rule

Let X be a random vector with density f_X with respect to the Lebesgue measure on $(\mathbb{R}^n, \mathcal{B}^n)$. Let $Z := g(X)$, where $g : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is invertible with inverse g^{-1} and Jacobian $|\frac{\partial z}{\partial x}|$. Then, at $z = g(x)$ the random vector Z has the following density with respect to the Lebesgue measure on $(\mathbb{R}^n, \mathcal{B}^n)$:

$$f_Z(z) := \frac{f_X(x)}{|\frac{\partial z}{\partial x}|} = f_X(g^{-1}(z)) \left| \frac{\partial x}{\partial z} \right|, \quad z \in \mathbb{R}^n. \quad (2.9)$$

Proposition 2.2: Expectation of a Quadratic Form

Let A be an $n \times n$ matrix and X an n -dimensional random vector with expectation vector μ and covariance matrix Σ . The random variable $Y := X^\top A X$ has expectation $\text{tr}(\Sigma A) + \mu^\top A \mu$.

2.5 L^p Spaces

Let X be a numerical random variable on

Definition 2.11: L^p Space

For each $p \in [1, \infty]$ the space L^p is comprised of all numerical random variables X for which $\|X\|_p < \infty$.

Lemma 2.2: Jensen's Inequality

Let $h : \mathcal{X} \rightarrow \mathbb{R}$ be a convex function and let X be a random variable taking values in $\mathcal{X}$, with expectation vector $\mathbb{E}X$. Then,

$$\mathbb{E}h(X) \geq h(\mathbb{E}X).$$

Theorem 2.8: Properties of the L^p Norm

Let X and Y be numerical random variables on $(\Omega, \mathcal{H}, \mathbb{P})$. Then, the following hold:

- (1) (Positivity): $\|X\|_p \geq 0$, and $\|X\|_p = 0 \iff X = 0$ almost surely.
- (2) (Multiplication by a constant): $\|cX\|_p = |c|\|X\|_p$.
- (3) (Minkowski's (triangle) inequality):

$$|\|X\|_p - \|Y\|_p| \leq \|X + Y\|_p \leq \|X\|_p + \|Y\|_p.$$

- (4) (Hölder's inequality): For $p, q, r \in [1, \infty]$ with $\frac{1}{p} + \frac{1}{q} = \frac{1}{r}$,

$$\|XY\|_r \leq \|X\|_p \|Y\|_q. \tag{2.10}$$

- (5) (Monotonicity): If $1 \leq p < q \leq \infty$, then $\|X\|_p \leq \|X\|_q$.

2.6 Integral Transforms

Expectations are used in many probabilistic analyses.

Definition 2.12: Moment Generating Function

The *moment generating function (MGF)* of a numerical random variable X with distribution μ is the function $M : \mathbb{R} \rightarrow [0, \infty]$, given by

$$M(s) := \mathbb{E}e^{sX} = \int_{-\infty}^{\infty} \mu(dx)e^{sx}, \quad s \in \mathbb{R}.$$

Theorem 2.9: Uniqueness of the MGF

A measure μ is determined by its MGF M if the latter is finite in a neighborhood of 0.

Theorem 2.10: Taylor's Theorem for MGFs

Suppose $M(s) := \mathbb{E}e^{sX}$ is finite in the neighborhood $(-s_0, s_0)$ of 0. Then, M has the Taylor expansion

$$M(s) = \sum_{k=0}^{\infty} \frac{s^k}{k!} \mathbb{E}X^k, \quad |s| < s_0.$$

In particular, all moments $\mathbb{E}X^k, k \in \mathbb{N}$ are finite.

Theorem 2.11: Product of MGFs

If X and Y are independent, then $M_{X+Y}(s) = M_X(s)M_Y(s)$.

Definition 2.13: Characteristic Function

The *characteristic function* of a numerical random variable X with distribution μ is the function $\psi : \mathbb{R} \rightarrow \mathbb{C}$, defined by

$$\psi(r) := \mathbb{E}e^{irX} = \mathbb{E} \cos(rX) + i\mathbb{E} \sin(rX) = \int_{-\infty}^{\infty} \mu(dx)e^{irx}, \quad r \in \mathbb{R}.$$

Theorem 2.12: Inversion and Uniqueness of the Characteristic Function

If the probability measure μ has characteristic function ψ , and if $\mu\{a\} = \mu\{b\} = 0$, then

$$\mu(a, b] = \lim_{t \rightarrow \infty} \frac{1}{2\pi} \int_{\mathbb{R}} dr \mathbb{1}_{[-t, t]}(r) \frac{e^{-ira} - e^{-irb}}{ir} \psi(r) := \lim_{t \rightarrow \infty} I_t. \quad (2.11)$$

In particular, the characteristic function determines μ uniquely. If, in addition, $\int_{\mathbb{R}} dr |\psi(r)| < \infty$, then μ has a density f with respect to the Lebesgue measure, given by

$$f(x) := \frac{1}{2\pi} \int_{-\infty}^{\infty} dr e^{-irx} \psi(r). \quad (2.12)$$

Proposition 2.3: Properties of the Characteristic Function

- (1) (Symmetry): The random variables X and $-X$ are identically distributed if and only if the characteristic function of X is real-valued.
- (2) (Convolution): If X and Y are independent, then $\psi_{X+Y} = \psi_X \psi_Y$.
- (3) (Affine transformation): If a and b are real numbers, then

$$\psi_{aX+b}(r) = e^{irb} \psi_X(ar), \quad r \in \mathbb{R}.$$

- (4) (Taylor's theorem): If $\mathbb{E}|X^n| < \infty$, then

$$\psi(r) = \sum_{k=0}^n \frac{\mathbb{E}X^k}{k!} (ir)^k + O(r^n), \quad r \in \mathbb{R}.$$

2.7 Information and Independence

In this section, we have another look at the role of σ -algebras and independence in random experiments.

Definition 2.14: σ -Algebra Generated by a Random Variable

Let X be a random variable taking values in $(E, \mathcal{E})$. The sets $\{X \in A\}, A \in \mathcal{E}$ form a σ -algebra, called the σ -algebra generated by X , and we write σX .

Theorem 2.13: σ -Algebra Generated by a Stochastic Process

For the stochastic process $X := \{X_t, t \in \mathbb{T}\}$, the σ -algebra generated by X is the smallest σ -algebra on Ω that is generated by the union of the σ -algebras $\sigma X_t, t \in \mathbb{T}$. We write $\bigvee_{t \in \mathbb{T}} \sigma X_t$ or $\sigma\{X_t, t \in \mathbb{T}\}$.

Theorem 2.14: σX -Numerical Random Variables

Let X be a random variable taking values in a measurable space $(E, \mathcal{E})$. A mapping $V : \Omega \rightarrow \bar{\mathbb{R}}$ belongs to σX if and only if $V = h \circ X$ for some $h \in \mathcal{E}$.

Definition 2.15: Filtration

Let $\mathbb{T}$ be a subset of $\mathbb{R}$. A *filtration* is an increasing collection $\{\mathcal{F}_t : t \in \mathbb{T}\}$ of sub- σ -algebras of $\mathcal{H}$; that is, $\mathcal{F}_s \subseteq \mathcal{F}_t$ whenever $s < t$.

Theorem 2.15: Stochastic Process with Arbitrary Index Set

For each t in an arbitrary set $\mathbb{T}$, let X_t be a random variable taking values in a measurable set $(E_t, \mathcal{E}_t)$. Then, the mapping $V : \Omega \rightarrow \bar{\mathbb{R}}$ belongs to $\sigma\{X_t, t \in \mathbb{T}\}$ if and only if there exists a sequence (t_n) in $\mathbb{T}$ and a function h in $\bigotimes_n \mathcal{E}_{t_n}$ such that

$$V = h(X_{t_1}, X_{t_2}, \dots).$$

Definition 2.16: Independence for σ -Algebras

Let $\{\mathcal{F}_i, i = 1, \dots, n\}$ be a collection of sub- σ -algebras of $\mathcal{H}$. The $\{\mathcal{F}_i\}$ are said to be (mutually) independent or form an *independency* if

$$\mathbb{E}V_1 \cdots V_n = \mathbb{E}V_1 \cdots \mathbb{E}V_n \quad (2.13)$$

for all positive random variables $V_i \in \mathcal{F}_i, i = 1, \dots, n$.

Theorem 2.16: Independence of σ -Algebras

Sub- σ -algebras $\mathcal{F}_1, \dots, \mathcal{F}_n$ of $\mathcal{H}$ generated by p -systems $\mathcal{C}_1, \dots, \mathcal{C}_n$ are independent if and only if for all $H_i \in \mathcal{C}_i \cup \{\Omega\}$, $i = 1, \dots, n$, it holds that:

$$\mathbb{P}(H_1 \cap \dots \cap H_n) = \mathbb{P}(H_1) \dots \mathbb{P}(H_n). \quad (2.14)$$

Theorem 2.17: Independence of Random Variables

Random variables X and Y taking values in $(E, \mathcal{E})$ and $(F, \mathcal{F})$ are independent if and only if

$$\mathbb{E}g(X)h(Y) = \mathbb{E}g(X)\mathbb{E}h(Y), \quad g \in \mathcal{E}_+, h \in \mathcal{F}_+. \quad (2.15)$$

2.8 Important Stochastic Processes

We conclude by highlighting some key classes of stochastic processes, including Gaussian processes, Poisson random measures, and Lévy processes.

Definition 2.17: Gaussian Process

A real-valued stochastic process $\{X_t, t \in \mathbb{T}\}$ is said to be *Gaussian* if all its finite-dimensional distributions are Gaussian (normal); that is, if the vector $[X_{t_1}, \dots, X_{t_n}]^\top$ is multivariate normal for any choice of n and $t_1, \dots, t_n \in \mathbb{T}$.

Definition 2.18: Poisson Random Measure

A random measure N on $(E, \mathcal{E})$ is said to be a *Poisson random measure* with mean measure μ if the following properties hold:

- (1) $N(A) \sim \text{Poi}(\mu(A))$ for any set $A \in \mathcal{E}$.
- (2) For any selection of disjoint sets $A_1, \dots, A_n \in \mathcal{E}$, the random variables $N(A_1), \dots, N(A_n)$ are independent.

Proposition 2.4: Poisson Random Measure on a Product Space

Let $X := \{X_i, i \in I\}$ be the atoms of a Poisson random measure N on $(E, \mathcal{E})$ with mean measure μ , and let $\{Y_i, i \in \mathbb{N}\}$ be i.i.d. random variables on $(F, \mathcal{F})$ with distribution π , independent of X . Then, $\{(X_i, Y_i), i \in I\}$ form the atoms of a Poisson random measure M on $(E \times F, \mathcal{E} \otimes \mathcal{F})$ with mean measure $\mu \otimes \pi$.

Theorem 2.18: Conditioning on the Total Number of Points

Let N be a Poisson random measure on $(E, \mathcal{E})$ with mean measure μ satisfying $\mu(E) := c < \infty$. Then, conditional on $N(E) = k$, the k atoms of N are i.i.d. with probability distribution μ/c .

Definition 2.19: Compound Poisson Process

The process $(X_t, t \geq 0)$ defined by

$$X_t := \int_{[0,t] \times \mathbb{R}^d} N(ds, dx) x, \quad t \geq 0,$$

is the *compound Poisson process* corresponding to the measure ν .

Definition 2.20: Lévy Process

The process X is said to be a *Lévy process* if

- (1) the paths of X are almost surely right-continuous and left-limited, with $X_0 = 0$,
and
- (2) for every $t_1 < t_2 < t_3 < t_4$, the increments $X_{t_2} - X_{t_1}$ and $X_{t_4} - X_{t_3}$ are independent,
and for every $t, u \in \mathbb{R}_+$, the increment $X_{t+u} - X_t$ has the same distribution as X_u .

Theorem 2.19: Lévy-Itô Decomposition

The process $X := (X_t, t \geq 0)$ is a Lévy process if and only if there exists a d -dimensional Brownian motion process $B := (B_t, t \geq 0)$ starting at $\mathbf{0}$ and a Poisson random measure N on $\mathbb{R}_+ \times \mathbb{R}^d$ that is independent of B and has mean measure $\text{Leb} \otimes \nu$, with ν satisfying

$$\int_{\mathbb{R}^d} \nu(d\mathbf{x}) (|\mathbf{x}|^2 \wedge 1) < \infty, \quad (2.16)$$

such that $X = B + U + Z$, where the summands are independent, $U := (U_t, t \geq 0)$ is the compound Poisson process

$$U_t := \int_0^t \int_{\|\mathbf{x}\| > 1} N(ds, d\mathbf{x}) \mathbf{x}, \quad t \geq 0,$$

and $Z := (Z_t, t \geq 0)$ is the limit of compensated compound Poisson processes:

$$Z := \lim_{\delta \downarrow 0} Z^\delta \quad \text{with} \quad Z_t^\delta := \int_0^t \int_{\delta \leq \|\mathbf{x}\| \leq 1} [\mathbf{x} N(ds, d\mathbf{x}) - ds \mathbf{x} \nu(d\mathbf{x})], \quad t \geq 0.$$

3 Convergence

4 Conditioning

5 Martingales

6 Wiener and Brownian Motion Processes

7 Itô Calculus

7.1 Itô Integral

We will work with a Wiener process W on a complete probability space $(\Omega, \mathcal{H}, \mathbb{P})$ with a right-continuous filtration $\mathcal{F} := (\mathcal{F}_s, s \geq 0)$ and such that W is adapted to $\mathcal{F}$. We also use the following definition.

Definition 7.1: Simple Processes

A process $F := (F_s, s \in [0, t])$ is called *simple* if it is of the form

$$F_s := \xi_0 \mathbb{1}_{\{0\}}(s) + \sum_{k=0}^{n-1} \xi_k \mathbb{1}_{(s_k, s_{k+1}]}(s), \quad s \in [0, t], \quad (7.1)$$

where $\{s_0, \dots, s_n : 0 = s_0 < \dots < s_n = t\} =: \Pi_n, n \in \mathbb{N}$, is a segmentation of $[0, t]$ and $\xi_k \in \mathcal{F}_{s_k}$ for all k .

Note that the simple process in [Equation \(7.1\)](#) is right-continuous at $s = 0$ and is left-continuous at $s \in (0, t]$. [Figure 2](#) shows a realisation of a simple process.

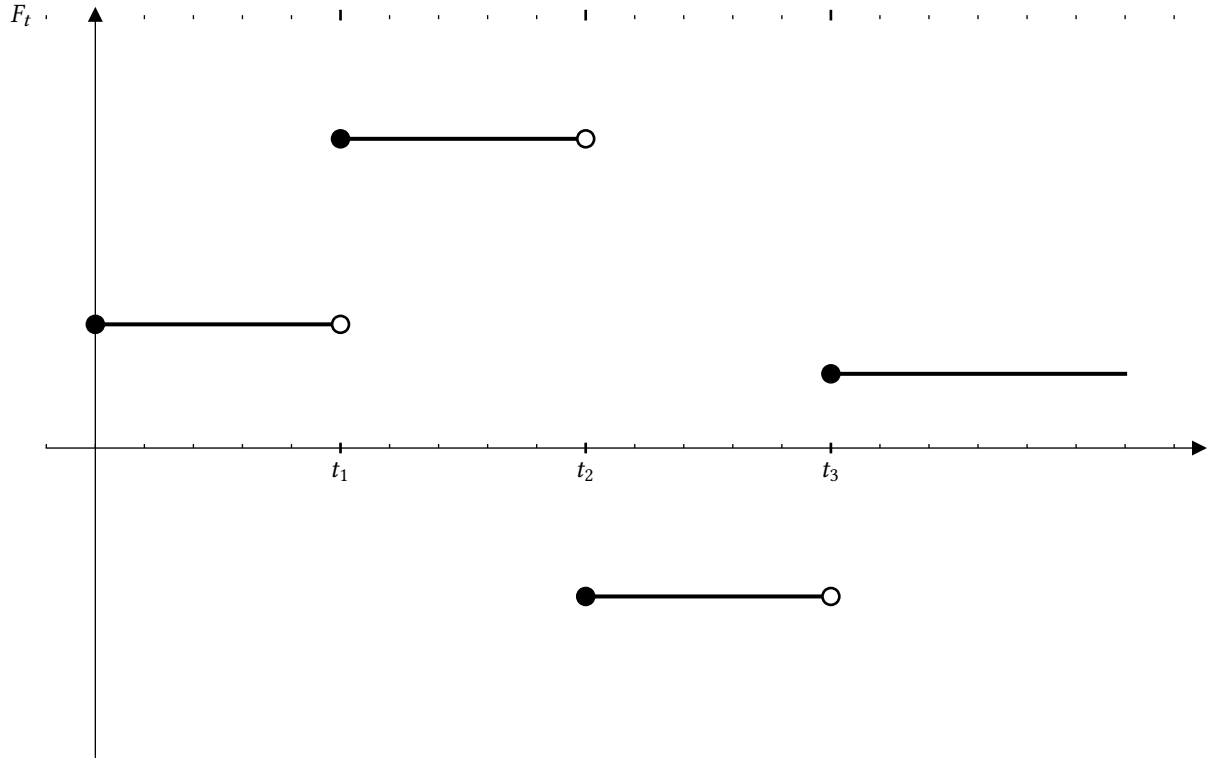


Figure 2: A path of a simple process.

We will consider the interplay between the simple process F_t and the Brownian motion $W(t)$ in the following way. Regard $W(t)$ as the price per share of an asset at time t . Since Brownian motion can take both negative and positive values, it is not a suitable model for the price of a limited-liability asset, such as a stock. Notwithstanding, for the sake of this illustration, we ignore that issue. Think of $t_0, t_1, \dots, t_{n-1}$ as the *trading dates* in the asset, and think of $F_{t_0}, F_{t_1}, \dots, F_{t_{n-1}}$ as the position (number of shares) taken in the asset at each trading date and held to the next trading date. The gain from trading at each time t is given by

$$I(t) = F_{t_0} [W(t) - W(t_0)] = F_0 W(t), \quad 0 \leq t \leq t_1, \quad (7.2)$$

$$I(t) = F_0 W(t_1) + F_{t_1} [W(t) - W(t_1)], \quad t_1 \leq t \leq t_2, \quad (7.3)$$

$$I(t) = F_0 W(t_1) + F_{t_1} [W(t_2) - W(t_1)] + F_{t_2} [W(t) - W(t_2)], \quad t_2 \leq t \leq t_3. \quad (7.4)$$

and so on. In general, if $t_k \leq t \leq t_{k+1}$, then

$$I(t) = \sum_{j=0}^{k-1} F_{t_j} [W(t_{j+1}) - W(t_j)] + F_{t_k} [W(t) - W(t_k)]. \quad (7.5)$$

We are now ready to define the Itô Integral for simple processes

Definition 7.2: Itô Integral for Simple Processes

Suppose that F is a simple process of the form Equation (7.1), with $\int_0^t \mathbb{E} F_s^2 ds < \infty$.

Then, the *Itô integral* of F with respect to W over $[0, r]$ is defined as

$$\int_0^r F_s dW_s := \sum_{k=0}^{n-1} \xi_k (W_{s_{k+1} \wedge r} - W_{s_k \wedge r}), \quad r \in [0, t]. \quad (7.6)$$

The Itô integral from r to t is naturally defined as the difference of two integrals:

$$\int_r^t F_s dW_s = \int_0^t F_s dW_s - \int_0^r F_s dW_s. \quad (7.7)$$

The Itô integral has the important *isometry* property:

$$\mathbb{E} \left[\left(\int_0^t F_s dW_s \right)^2 \right] = \int_0^t \mathbb{E} [F_s^2] ds, \quad (7.8)$$

which allows us to compute the variance of the Itô integral. In fact, the Itô integral satisfies the following four important properties:

Theorem 7.1: Properties of the Itô Integral

Assume that the Itô integral is well-defined. Then, for $\alpha, \beta \in \mathbb{R}$ and $r \in [0, t]$:

- (1) (Linearity): $\int_0^r [\alpha F_s + \beta G_s] dW_s = \alpha \int_0^r F_s dW_s + \beta \int_0^r G_s dW_s$.
- (2) (Zero mean): $\mathbb{E} \int_0^r F_s dW_s = 0$.
- (3) (Isometry): $\mathbb{E} \left[\int_0^r F_s dW_s \times \int_0^r G_s dW_s \right] = \int_0^r \mathbb{E} [F_s G_s] ds$.
- (4) (Martingale): $(\int_0^r F_s dW_s, r \in [0, t])$ is an L^2 -martingale.

Definition 7.3: Canonical Approximation

For any $\mathcal{F}$ -adapted process $F := (F_s, s \in [0, t])$, we define its *canonical approximation* as

$$F_s^{(n)} := F_0 \mathbb{1}_{\{0\}}(s) + \sum_{k=0}^{n-1} F_{s_k} \mathbb{1}_{(s_k, s_{k+1}]}(s), \quad s \in [0, t]. \quad (7.9)$$

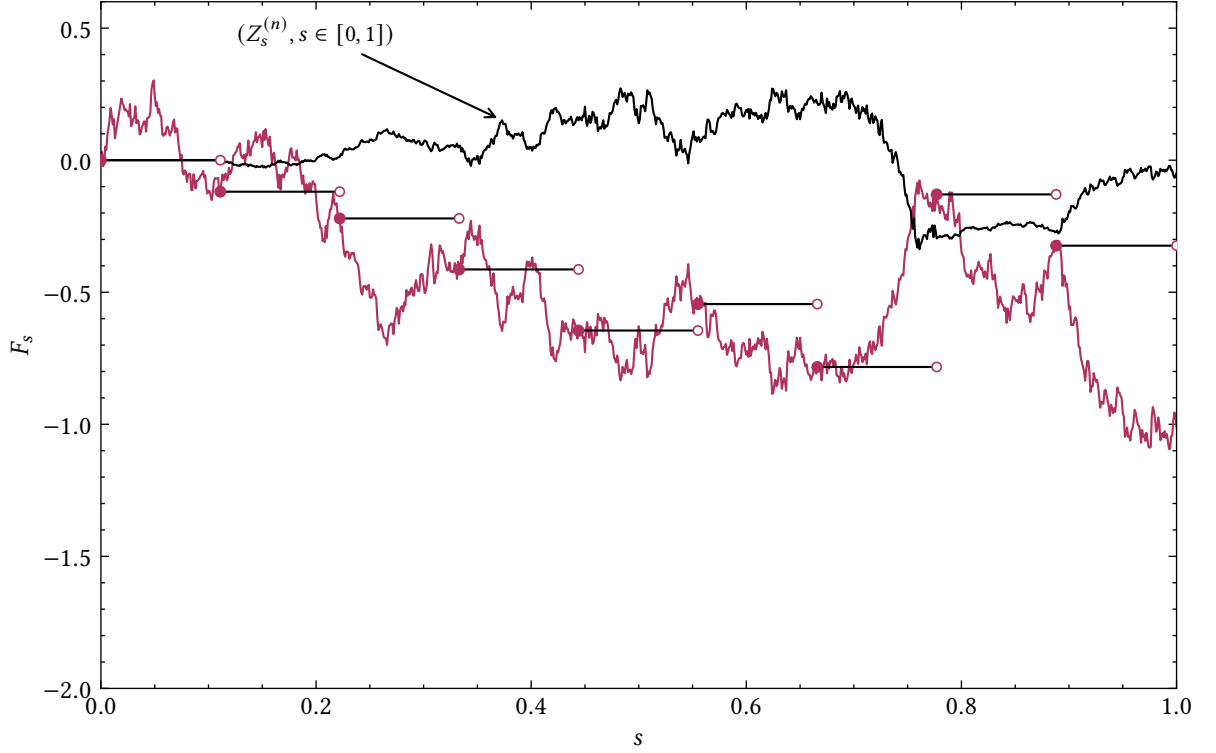


Figure 3: A Wiener process $F := (W_s, s \in [0, t])$, its canonical approximation $F^{(n)}$ and the Itô integral process $Z^{(n)} = (\int_0^r F_s^{(n)} dW_s, r \in [0, 1])$

Having defined the Itô integral for simple processes, we are now ready to extend the definition to more general predictable processes.

7.1.1 Itô Integral for Predictable Processes

While simple processes of the form Equation (7.1) are $\mathcal{F}$ -predictable, it is not immediately clear if more general processes are also $\mathcal{F}$ -predictable. The following result shows that any process $F := (F_s, s \in [0, t])$ adapted to $\mathcal{F}$ and left-continuous on $(0, t]$ is not only $\mathcal{F}$ -predictable, but also *indistinguishable* from any other left-continuous modification $\tilde{F}$ of F ; that is, if $\mathbb{P}(F_s = \tilde{F}_s) = 1$ for all $s \in [0, t]$, then $\mathbb{P}(F_s = \tilde{F}_s \text{ for all } s \in [0, t]) = 1$. Moreover,

F is *progressively measurable*; that is, the mapping $F : [0, t] \times \Omega \rightarrow \mathbb{R}$ is measurable with respect to the product σ -algebra $\mathcal{B}_{[0,t]} \otimes \mathcal{F}_t$.

Theorem 7.2: Properties of Left-continuous Adapted Processes

Suppose that $F := (F_s, s \in [0, t])$ is a process on $(\Omega, \mathcal{H}, \mathbb{P})$ that is adapted to the filtration $\mathcal{F}$ and is left-continuous on $(0, t]$. Then, the following hold:

- (1) (Predictable): The process F is $\mathcal{F}$ -predictable (i.e., $\mathcal{F}^P$ -measurable).
- (2) (Progressively Measurable): For each t , the mapping $F : [0, t] \times \Omega \rightarrow \mathbb{R}$ is jointly measurable with respect to $\mathcal{B}_{[0,t]} \otimes \mathcal{F}_t$.
- (3) (Indistinguishable): If $\tilde{F}$ is a modification of F and almost surely left-continuous, then $\mathbb{P}(\cap_{s \leq t} \{F_s = \tilde{F}_s\}) = 1$.

Proof. Consider the canonical approximation $F^{(n)}$ of F , given in (7.9), where the segmentation Π_n satisfies $\text{mesh}(\Pi_n) \rightarrow 0$.

For $s = 0$ and $\omega \in \Omega$, we have $F_0(\omega) = F_0^{(n)}(\omega)$ by construction. For any $s \in (0, t]$, choose k (depending on n and s) with $s_k < s \leq s_{k+1}$. Define

$$\varepsilon_{n,s} := s - s_k, \quad 0 < \varepsilon_{n,s} \leq \text{mesh}(\Pi_n). \quad (7.10)$$

Then

$$F_s - F_s^{(n)} = F_s - F_{s_k} + \underbrace{F_{s_k} - F_s^{(n)}}_{=0} = F_s - F_{s-\varepsilon_{n,s}}, \quad (7.11)$$

and the last difference vanishes as $n \uparrow \infty$ because F is left-continuous. Hence, $F_s^{(n)}(\omega) \rightarrow F_s(\omega)$ for all $s \in [0, t]$ and $\omega \in \Omega$.

Since F is adapted to $\mathcal{F}$, Example 7.4 implies that each simple process $F^{(n)}$ is $\mathcal{F}$ -predictable. Consequently, for each t , the mapping $F^{(n)} : [0, t] \times \Omega \rightarrow \mathbb{R}$ is jointly measurable with respect to $\mathcal{B}_{[0,t]} \otimes \mathcal{F}_t$. Because pointwise limits of measurable functions are measurable, the same properties hold for the limit F . This establishes $\mathcal{F}$ -predictability and progressive measurability for F .

Now assume that $\widetilde{F}$ is a modification of F ; that is, $\mathbb{P}(\widetilde{F}_s \neq F_s) = 0$ for all $s \in [0, t]$. Let

$$B := \{\text{both } F \text{ and } \widetilde{F} \text{ are left-continuous on } (0, t]\}, \quad (7.12)$$

so that $\mathbb{P}(B) = 1$. Let $\mathbb{Q}_t := [0, t] \cap \mathbb{Q}$ and define

$$A := \bigcap_{s \in \mathbb{Q}_t} \{\widetilde{F}_s = F_s\}. \quad (7.13)$$

Since $\mathbb{Q}_t$ is countable, by countable subadditivity [Theorem 2.1](#),

$$\mathbb{P}(A^c) = \mathbb{P}\left(\bigcup_{s \in \mathbb{Q}_t} \{\widetilde{F}_s \neq F_s\}\right) \leq \sum_{s \in \mathbb{Q}_t} \mathbb{P}(\widetilde{F}_s \neq F_s) = 0, \quad (7.14)$$

so $\mathbb{P}(A) = 1$.

By denseness of $\mathbb{Q}_t$, for any $s \in [0, t]$ there exists a sequence $(s_n) \subset \mathbb{Q}_t$ with $s_n \uparrow s$. Therefore, for all $\omega \in A \cap B$,

$$F_s(\omega) - \widetilde{F}_s(\omega) = \lim_{n \uparrow \infty} (F_{s_n}(\omega) - \widetilde{F}_{s_n}(\omega)) = 0, \quad \forall s \in [0, t]. \quad (7.15)$$

Since $\mathbb{P}(A \cap B) = 1$, we conclude that F and $\widetilde{F}$ are indistinguishable. $\square$

We have so far considered stochastic processes that are left-continuous and adapted to the filtration $\mathcal{F}$. However, in order to define the Itô integral for these more general processes, additional regularity conditions are needed.

Definition 7.4: The Class of Processes $\mathcal{P}_t$

Let $\mathcal{P}_t$ be the class of stochastic processes $F := (F_s, s \in [0, t])$ that are left-continuous on $(0, t]$, $\mathcal{F}$ -adapted, and satisfy the norm condition:

$$\|F\|_{\mathcal{P}_t}^2 := \mathbb{E} \int_0^t F_s^2 ds < \infty.$$

Theorem 7.3: Approximating the Class $\mathcal{P}_t$ via Simple Processes

Suppose that $F \in \mathcal{P}_t$. Then, there exists a sequence $(F^{(n)})$ of simple processes of the form Equation (7.1) such that $\text{mesh}(\Pi_n) \rightarrow 0$ implies that $F_s^{(n)} \xrightarrow{a.s.} F_s$ for all $s \in [0, t]$ and

$$\lim_{n \rightarrow \infty} \|F^{(n)} - F\|_{\mathcal{P}_t} = 0. \quad (7.16)$$

Proposition 7.1: Deterministic Integrands and Gaussian Processes

Let $f : \mathbb{R}_+ \times \mathbb{R}_+ \rightarrow \mathbb{R}$ be left-continuous in the first argument and such that $\|f(\cdot, t)\|_{\mathcal{P}_t}^2 = \int_0^t f^2(s, t) ds < \infty$. Then, the stochastic integral process

$$Z_t := \int_0^t f(s, t) dW_s, \quad t \geq 0, \quad (7.17)$$

is a zero-mean Gaussian process with covariance function

$$\text{Cov}(Z_{t_1}, Z_{t_2}) = \int_0^{t_1 \wedge t_2} f(s, t_1) f(s, t_2) ds, \quad t_1, t_2 \geq 0. \quad (7.18)$$

$$\lim_{m \rightarrow \infty} \sup_{s \in [0, t]} \mathbb{E} F_s^2 \mathbf{1}_{\{F_s^2 > m\}} = 0. \quad (7.19)$$

Lemma 7.1: Quadratic Variation Integral

Suppose that $F \in \mathcal{P}_t$ satisfies the condition Equation (7.19). Then, as $\text{mesh}(\Pi_n) \rightarrow 0$:

$$\sum_{k=0}^{n-1} F_{s_k} (W_{s_{k+1}} - W_{s_k})^2 \xrightarrow{L^2} \int_0^t F_s ds. \quad (7.20)$$

Theorem 7.4: Enlarging the Class $\mathcal{P}_t$ via Localization

Let $F := (F_s, s \in [0, t])$ be an $\mathcal{F}$ -adapted process, left-continuous on $(0, t]$, and such that (7.34) holds. Then, there exists a localizing sequence (T_n) such that the stopped process $(\int_0^{r \wedge T_n} F_s dW_s, r \in [0, t])$ is a square-integrable martingale. In addition, the Itô integral of F with respect to W on $[0, t]$ is defined as the almost sure limit:

$$\int_0^t F_s dW_s := \lim_{n \rightarrow \infty} \int_0^{t \wedge T_n} F_s dW_s. \quad (7.21)$$

Proposition 7.2: Continuous Modification and Stopping Times

If $F \in \mathcal{P}_t$ and $T \leq t$ is a stopping time with respect to $\mathcal{F}$, then almost surely:

$$\int_0^T F_s dW_s = \int_0^t \mathbb{1}_{[0, T]}(s) F_s dW_s. \quad (7.22)$$

Theorem 7.5: Continuous Modification of the Itô Integral

Suppose that $F \in \mathcal{P}_t$ and $(F^{(n)})$ is a sequence of simple processes satisfying (7.17). Denote the limit $\int_0^r F_s^{(n)} dW_s =: Z_r^{(n)} \xrightarrow{L^2} Z_r$ as the Itô integral on $[0, r]$ for $r \leq t$. Then, there exists a modification $\tilde{Z}$ of $(Z_r, r \in [0, t])$ such that the mapping $r \mapsto \tilde{Z}_r$ is almost surely continuous. Moreover, this modification is unique up to indistinguishability.

Theorem 7.6: Existence of the Itô Integral with Integrand $F \in \mathcal{P}_t$

Suppose that $F \in \mathcal{P}_t$ and $(F^{(n)})$ is a sequence of simple processes satisfying (7.17). Then, the following limit exists in L^2 norm:

$$\int_0^t F_s dW_s := \lim_{n \rightarrow \infty} \int_0^t F_s^{(n)} dW_s, \quad r \in [0, t], \quad (7.23)$$

and it defines the Itô integral of F with respect to W on $[0, t]$.

7.2 Itô Calculus

Theorem 7.7: Itô Formula for the Wiener Process

Let W be a Wiener process and $f : \mathbb{R} \rightarrow \mathbb{R}$ be twice continuously differentiable with derivatives f' and f'' . Then, almost surely

$$f(W_t) - f(W_0) = \int_0^t f'(W_s) dW_s + \frac{1}{2} \int_0^t f''(W_s) ds.$$

Definition 7.5: Integral With Respect To An Itô Process

Let X be an Itô(μ, σ) process and F a left-continuous $\mathcal{F}$ -adapted process such that almost surely $\int_0^t (|F_s \mu_s| + (F_s \sigma_s)^2) ds < \infty$ for all t . Then, the integral of F with respect to X is defined as the Itô($F\mu, F\sigma$) process with:

$$\int_0^t F_s dX_s := \int_0^t F_s \mu_s ds + \int_0^t F_s \sigma_s dW_s. \quad (7.24)$$

Definition 7.6: Quadratic Variation On $[0, t]$

Let $\Pi_n := \{s_k, k = 0, \dots, n\}$ be a segmentation of $[0, t]$ and let $(X^{(n)})$ be a sequence of simple processes with almost sure limit $X_s := \lim_n X_s^{(n)}$ for all $s \in [0, t]$ as $\text{mesh}(\Pi_n) \rightarrow 0$. The *quadratic variation* of $X := (X_s, s \in [0, t])$ over $[0, t]$, denoted as $\langle X \rangle_t$, is defined as the limit in probability of

$$\langle X^{(n)} \rangle_t := \sum_{k=0}^{n-1} (X_{s_{k+1}}^{(n)} - X_{s_k}^{(n)})^2. \quad (7.25)$$

Theorem 7.8: Quadratic Variation Of An Itô Process

If X is an Itô(μ, σ) process, then

$$\langle X^{(n)} \rangle_t \xrightarrow{L^2} \langle X \rangle_t = \int_0^t \sigma_s^2 ds. \quad (7.26)$$

Furthermore, if $F \in \mathcal{P}_t$ satisfies the condition (7.19) and $\text{mesh}(\Pi_n) \rightarrow 0$, then:

$$\sum_{k=0}^{n-1} F_{s_k} (X_{s_{k+1}} - X_{s_k})^2 \xrightarrow{L^2} \int_0^t F_s \sigma_s^2 ds. \quad (7.27)$$

Theorem 7.9: Itô's Formula For Itô Processes

Let X be an Itô process and let f be twice continuously differentiable. Then,

$$f(X_t) = f(X_0) + \int_0^t f'(X_s) dX_s + \frac{1}{2} \int_0^t f''(X_s) d\langle X \rangle_s. \quad (7.28)$$

Definition 7.7: Covariation Of Two Itô Processes

Let X and Y be two Itô processes adapted to the same filtration $\mathcal{F}$. Then, the *covariation* between X and Y is defined as the quantity:

$$\langle X, Y \rangle_t := \frac{\langle X + Y \rangle_t - \langle X \rangle_t - \langle Y \rangle_t}{2}. \quad (7.29)$$

Theorem 7.10: Itô's Formula In $\mathbb{R}^d$

Let (X_t) be a d -dimensional Itô process, and let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be twice continuously differentiable in all variables. Then,

$$df(X_t) = \sum_{i=1}^d \partial_i f(X_t) dX_{t,i} + \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^d \partial_{ij} f(X_t) d\langle X_{\cdot,i}, X_{\cdot,j} \rangle_t. \quad (7.30)$$

Theorem 7.11: Itô Processes And Martingales

An Itô process in $\mathbb{R}^d$ is a local martingale if and only if its drift is 0.

7.3 Stochastic Differential Equations

Stochastic differential equations are based on the same principle as ordinary differential equations, relating an unknown function to its derivatives, but with the additional feature that part of the unknown function is driven by randomness.

Definition 7.8: Stochastic Differential Equation

A *stochastic differential equation* (SDE) for a stochastic process $(X_t, t \geq 0)$ is an expression of the form

$$dX_t = a(X_t, t)dt + b(X_t, t)dW_t, \quad (7.31)$$

where $(W_t, t \geq 0)$ is a Wiener process and a and b are deterministic functions, called the *drift* and *diffusion* functions. The resulting process $(X_t, t \geq 0)$ is called an *(Itô) diffusion*.

Consider the autonomous SDE

$$dX_t = a(X_t)dt + b(X_t)dW_t. \quad (7.32)$$

As already mentioned, the meaning of this SDE is that X is a stochastic process that satisfies the integral equation

$$X_t = X_0 + \int_0^t a(X_s)ds + \int_0^t b(X_s)dW_s. \quad (7.33)$$

However, it is not clear that such a process indeed exists. We show in this section that under mild conditions on a and b we can obtain a unique solution X of the SDE via a method of successive approximations, similar to *Picard iteration* for ordinary differential equations.

The condition that is placed on a and b is that they be *Lipschitz continuous*; that is, for all $x \in \mathbb{R}$ there exists a constant c such that

$$|a(y) - a(x)| + |b(y) - b(x)| \leq c|y - x|. \quad (7.34)$$

The successive approximation idea is very simple. Suppose that $X_0 = x$. Define $X_t^{(0)} := x$ for all t , and let

$$X_t^{(n+1)} := x + \int_0^t a(X_s^{(n)}) ds + \int_0^t b(X_s^{(n)}) dW_s \quad (7.35)$$

for $n \in \mathbb{N}$. The following is the main existence result for SDEs:

Theorem 7.12: Existence of SDE Solutions

For any $t \in [0, v]$, the sequence $(X^{(n)})$ in (7.35) converges almost surely in the uniform norm on $[0, t]$ to a process X that satisfies the integral equation (7.33).

Lemma 7.2: Integral Bound

Let $X^{(n)}, n = 1, 2, \dots$ be defined in (7.35), with $X^{(0)} = x$. Then, for every $v > 0$ there exists a constant α such that

$$\mathbb{E} \sup_{s \leq t} \left(X_s^{(n+1)} - X_s^{(n)} \right)^2 \leq \alpha \|X^{(n)} - X^{(n-1)}\|_{\mathcal{F}_t}^2, \quad t \leq v.$$

Definition 7.9: Itô Stratonovich Integral

Let X and Y be Itô processes such that $\int_0^t Y_s dX_s$ and $\int_0^t X_s dY_s$ are well-defined stochastic integrals. Then, the *Itô Stratonovich integral* of Y with respect to X is defined as

$$\int_0^t Y_s \circ dX_s := \int_0^t Y_s dX_s + \frac{1}{2} \langle X, Y \rangle_t. \quad (7.36)$$

Proposition 7.3: Itô Stratonovich Formula

Let (X_t) be a d -dimensional Itô process, and let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be three times continuously differentiable in all variables. Then,

$$df(X_t) = \sum_{i=1}^d \partial_i f(X_t) \circ dX_{t,i}. \quad (7.37)$$

$$\tilde{\mathbb{P}}^t(A) := \mathbb{E} M_t 1_A, \quad A \in \mathcal{F}_t. \quad (7.38)$$

$$M_t = \exp \left(\int_0^t \mu_s dW_s - \frac{1}{2} \int_0^t \mu_s^2 ds \right) = e^{Z_t - \frac{1}{2} \langle Z \rangle_t}, \quad t \geq 0, \quad (7.39)$$

$$\lim_{\varepsilon \downarrow 0} \varepsilon \ln \mathbb{E} \exp((1 - \varepsilon)\langle Z \rangle_t / 2) = 0. \quad (7.40)$$

7.3.1 Methods for Solving Simple SDEs

If a strong solution of the Itô diffusion SDE exists, then it can be expressed as $X_t = \phi(X_0, (W_s, s \in [0, t]), t)$ for some function ϕ . Finding an analytical solution for ϕ is usually impossible, however, except in a few special cases.

7.3.1.1 Linear Stochastic Differential Equations One setting where the strong solution of an SDE can be expressed as a simple analytical formula is when both the drift and diffusion of the SDE are linear functions of the state variables. In particular, suppose that

$$a(x, t) = \alpha_t x + \beta_t,$$

$$b(x, t) = \delta_t x + \gamma_t.$$

An SDE of the form

$$dX_t = (\alpha_t + \beta_t X_t)dt + (\delta_t + \gamma_t X_t)dW_t \quad (7.41)$$

is said to be *linear*.

The solution to the general linear SDE in Equation (7.41) is given next.

Proposition 7.4: Solution to a Linear Diffusion SDE

The solution to the linear SDE Equation (7.41) is given by

$$X_t = Y_t \left(X_0 + \int_0^t \frac{\alpha_s - \delta_s \gamma_s}{Y_s} ds + \int_0^t \frac{\delta_s}{Y_s} dW_s \right). \quad (7.42)$$

Where

$$Y_t = \exp \left(\int_0^t \left(\beta_s - \frac{\gamma_s^2}{2} \right) ds + \int_0^t \gamma_s dW_s \right) \quad (7.43)$$

is the stochastic exponential of an Itô(β, γ) process.

Theorem 7.13: Girsanov's Change of Measure

Suppose that μ satisfies the Krylov condition [Equation \(7.40\)](#) and M is the exponential martingale [Equation \(7.39\)](#). Then, under the change of measure [Equation \(7.38\)](#) induced by M , the Itô process

$$\widetilde{W}_t := - \int_0^t \mu_s ds + W_t, \quad t \geq 0$$

is a Wiener process.

A fundamental stochastic process in finance is defined as the solution to the SDE

GBM.py

```

1  # Generate Brownian Motion
2  t = np.linspace(0, T, N + 1)
3  W = np.r_[0, np.cumsum(rng.normal(0, np.sqrt(T/N), N))]
4
5  # Exact Solution
6  S_t = S_0 * np.exp((mu - 0.5 * sigma**2) * t + sigma * W)

```

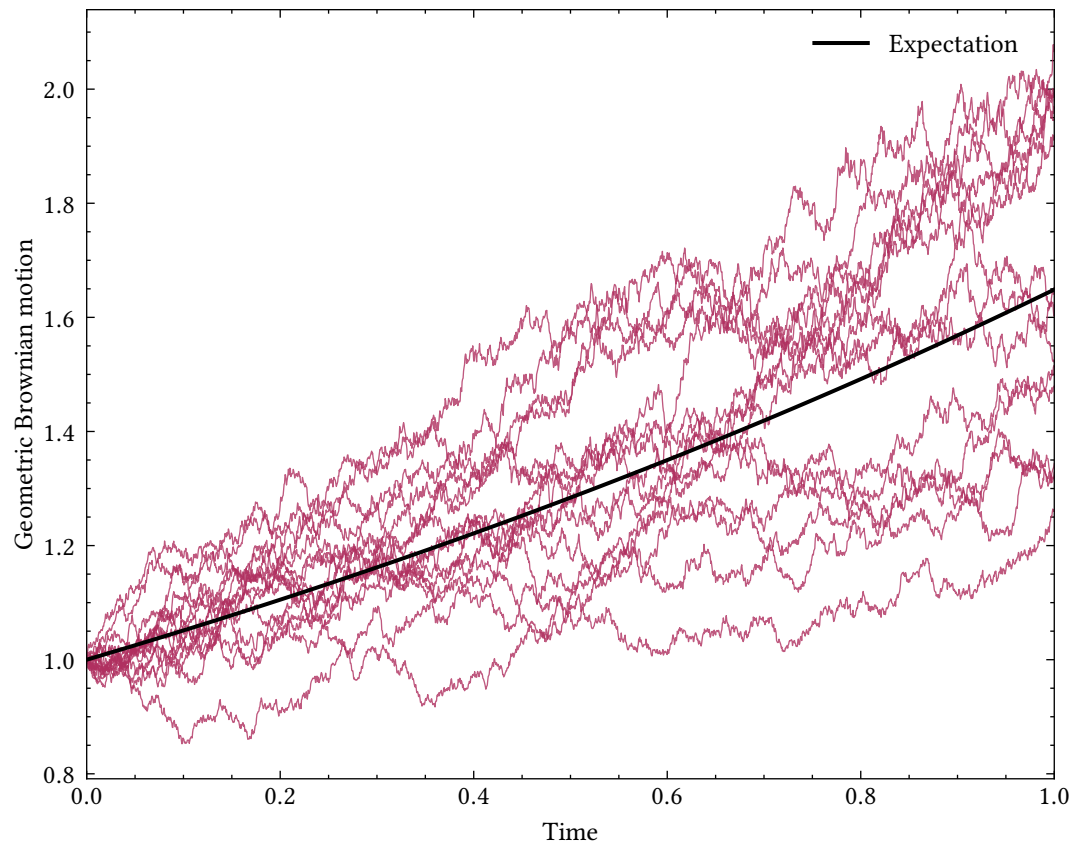


Figure 4: Geometric Brownian motion sample path and expectation for $\mu = 0.5$, $\sigma = 0.2$ and $S_0 = 1$.

Part II

Levy Processes

Part III

Stochastic Volatility

Part IV

Risk-Neutral Valuation

8 Discrete-time Market Model

9 Continuous-time Market Model

Part V

Martingale Methods

10 The Black-Scholes Model

11 A Dividend-paying Stock

12 Discretisation Schemes

13 Connection to Partial Differential Equations

Part VI

Monte Carlo Methods

SIMULATION techniques are invaluable in managing complex financial scenarios that involve path-dependent payoffs or distributions without analytical expressions. One prominent application is pricing derivative securities, estimating value at risk, and emulating hedging strategies. Monte Carlo simulation, in particular, proves highly effective for such tasks, offering a blend of simplicity and adaptability as its primary advantages. However, the method's drawback lies in its considerable computational demands, which can be mitigated to some extent. Our approach employs Monte Carlo simulation to generate simulated option price values, which are then compared against outcomes derived from analytical formulas.

14 Generating Random Numbers and Random Variables

15 Quasi Monte Carlo

It is customary in this setting to focus on the problem of numerical integration over the unit hypercube.

The digital constructions are similar to feedback shift register random number generators, while lattice rules are similar to congruential generators. One useful way to think of QMC is that we are utilising a small random number generator in its entirety. The tools used to develop and analyse low-discrepancy methods are very different from those used in ordinary Monte Carlo, as they draw on number theory and abstract algebra rather than probability and statistics.

15.1 Discrepancy Measures

15.2 Low-discrepancy Sequences

15.2.1 Halton and Hammersley

15.2.2 Faure

15.2.3 Sobol

15.3 Koksma-Hlawka

16 Variance Reduction

Part VII

Grid & Lattice Methods

17 Lattice Methods

18 Finite Difference Methods

19 Contract Sensitivities

19.1 Likelihood Ratio

19.2 Pathwise Derivatives

19.3 Finite-Difference Approximations

Part VIII

Financial Derivatives

20 European Options

21 Exotic Options

21.1 Introduction

The European calls and puts considered thus far are sometimes called *vanilla* or even *plain vanilla* options. Their payoffs depend only on the final value of the underlying asset. Options whose payoffs depend on the path of the underlying asset are called *path-dependent* or *exotic*.

We present three types of exotic options on a geometric Brownian motion asset and work out a detailed analysis for one option of each type. The types considered are *barrier options*, *lookback options*, and *Asian options*. We work out the standard partial differential equations governing the option price. The first two options have explicit pricing formulas, which are based on the reflection principle for Brownian motion. Such a formula for Asian options is not known. However, for the Asian option there is a change-of-numéraire argument that

reduces the pricing partial differential equation to a simple form that can easily be solved numerically.

21.2 Maximum of Brownian Motion with Drift

We derive the joint density for a Brownian motion with drift and its maximum to date. This density is used to obtain explicit pricing formulas for a barrier option and a lookback option.

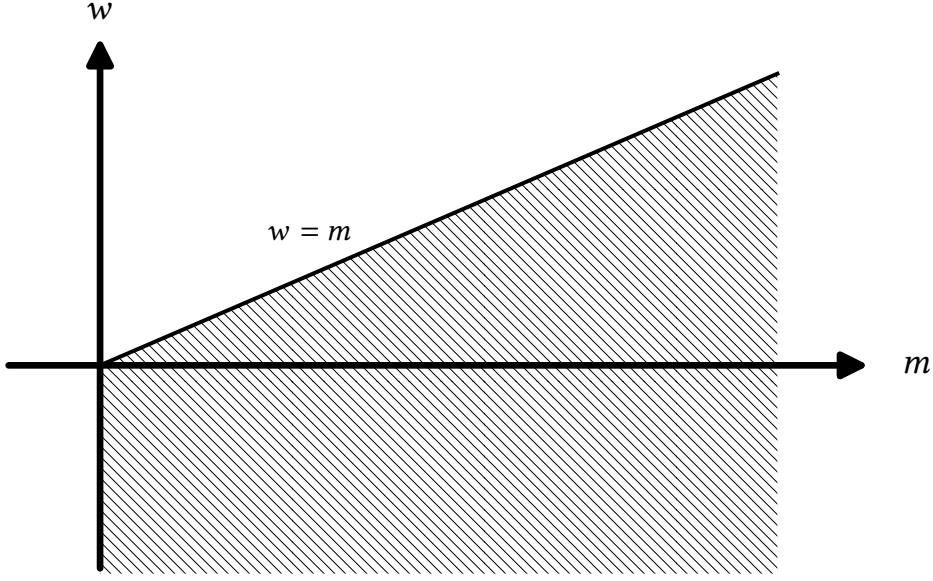
To derive this formula, we begin with a Brownian motion $\tilde{W}(t)$, $0 \leq t \leq T$, defined on a probability space $(\Omega, \mathcal{F}, \tilde{\mathbb{P}})$. Under $\tilde{\mathbb{P}}$, the Brownian motion $\tilde{W}(t)$ has zero drift (i.e., it is a martingale). Let α be a given number, and define

$$\hat{W}(t) = \alpha t + \tilde{W}(t), \quad 0 \leq t \leq T. \quad (21.1)$$

This Brownian motion $\hat{W}(t)$ has drift α under $\tilde{\mathbb{P}}$. We further define

$$\hat{M}(T) = \max_{0 \leq t \leq T} \hat{W}(t). \quad (21.2)$$

Because $\hat{W}(0) = 0$, we have $\hat{M}(T) \geq 0$. We also have $\hat{W}(T) \leq \hat{M}(T)$. Therefore, the pair of random variables $(\hat{M}(t), \hat{W}(T))$ takes values in the set $\{(m, w); w \leq m, m \geq 0\}$ shown in Figure 5.

Figure 5: Range of the pair $(\widehat{M}(T), \widehat{W}(T))$.**Theorem 21.1: Joint Density Under $\widetilde{\mathbb{P}}$**

The joint density under $\widetilde{\mathbb{P}}$ of the pair $(\widehat{M}(T), \widehat{W}(T))$ is

$$\widetilde{f}_{\widehat{M}(T), \widehat{W}(T)}(m, w) = \frac{2(2m - w)}{T\sqrt{2\pi T}} e^{\alpha w - \frac{1}{2}\alpha^2 T - \frac{1}{2T}(2m - w)^2}, \quad w \leq m, m \geq 0, \quad (21.3)$$

and is zero for other values of m and w .

Corollary 21.1: Marginal Distribution and Density

We have

$$\widetilde{\mathbb{P}}\{\widehat{M}(T) \leq m\} = N\left(\frac{m - \alpha T}{\sqrt{T}}\right) - e^{2\alpha m} N\left(\frac{-m - \alpha T}{\sqrt{T}}\right), \quad m \geq 0, \quad (21.4)$$

and the density under $\widetilde{\mathbb{P}}$ of the random variable $\widehat{M}(T)$ is

$$\widetilde{f}_{\widehat{M}(T)}(m) = \frac{2}{\sqrt{2\pi T}} e^{-\frac{1}{2T}(m - \alpha T)^2} - 2\alpha e^{2\alpha m} N\left(\frac{-m - \alpha T}{\sqrt{T}}\right), \quad m \geq 0, \quad (21.5)$$

and is zero for $m < 0$.

Figure 6 represents the running maximum process

$$M_0^t := \max_{s \in [0, t]} W_s, \quad t \geq 0,$$

of Brownian motion $(W_t)_{t \in \mathbb{R}_+}$.

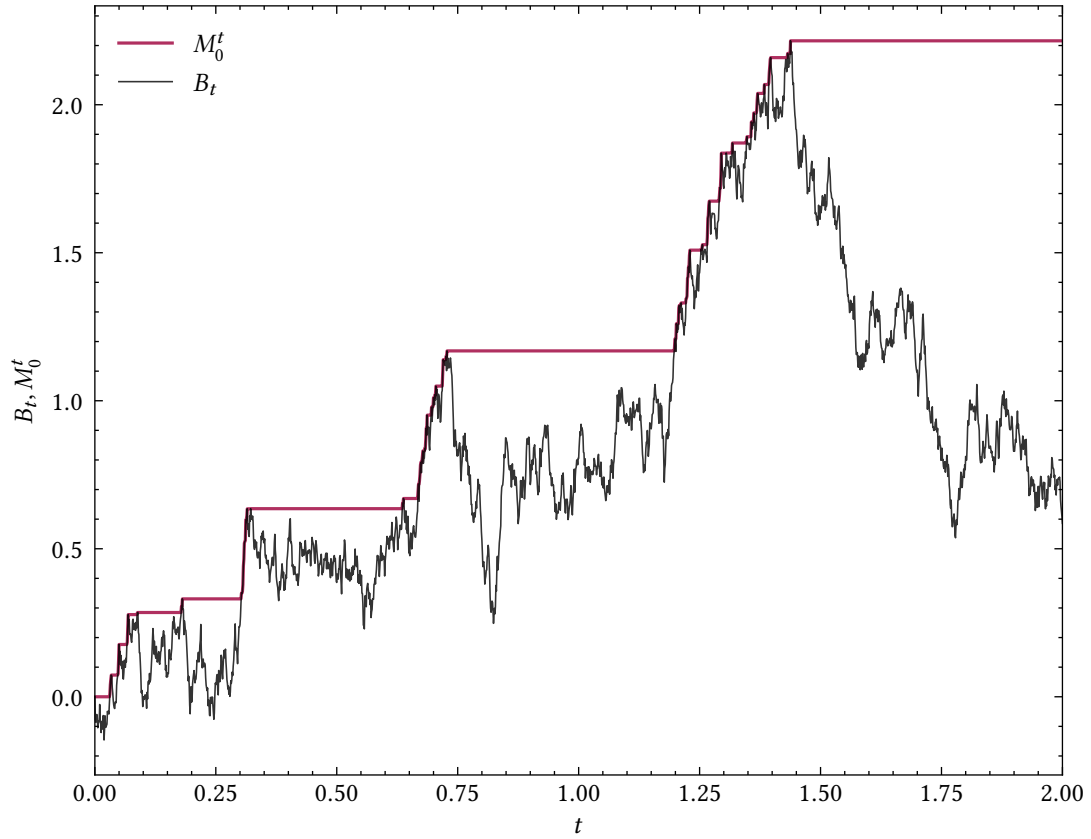


Figure 6: A Wiener process path B_t and its running maximum M_0^t .

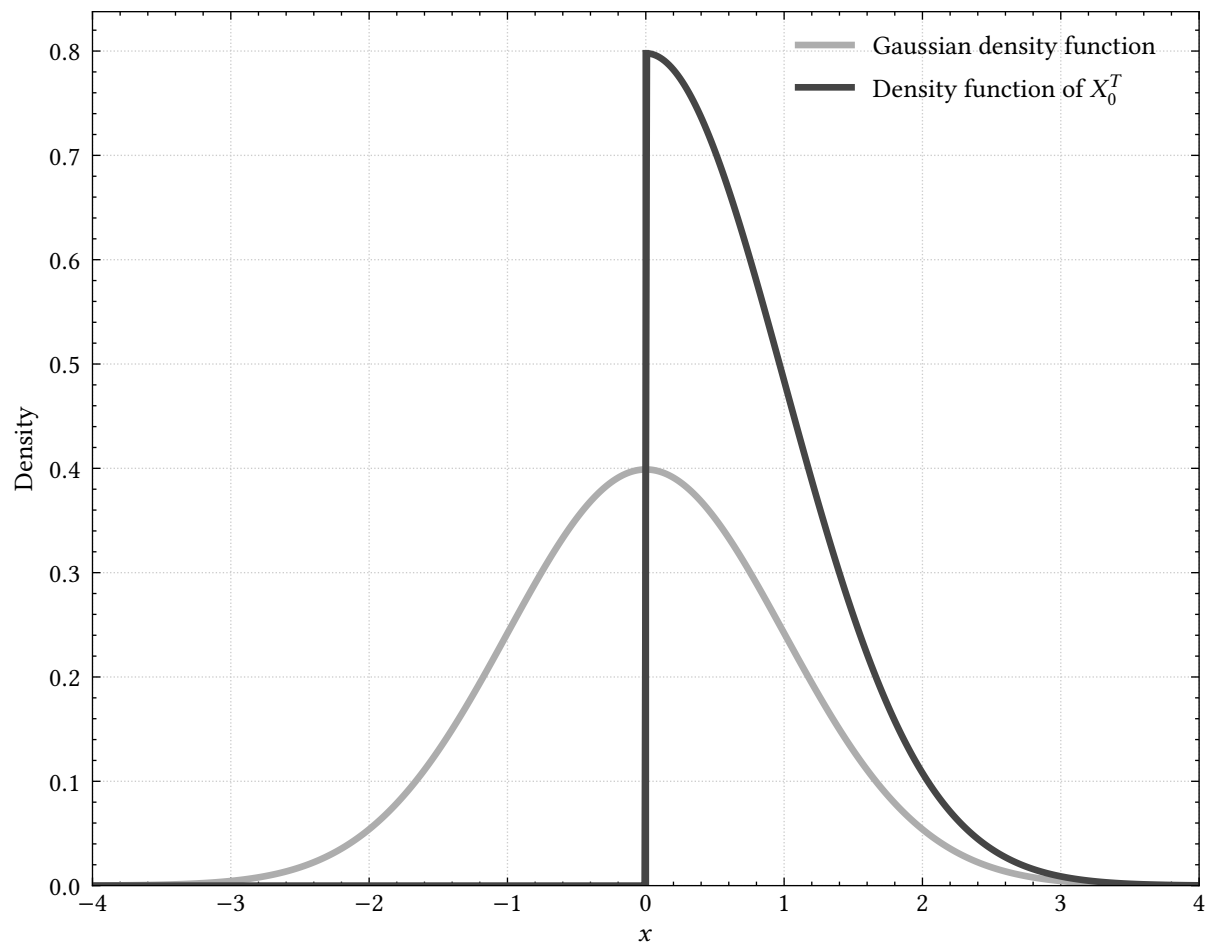


Figure 7: Probability density of the maximum X_0^1 of Brownian motion over $[0, 1]$

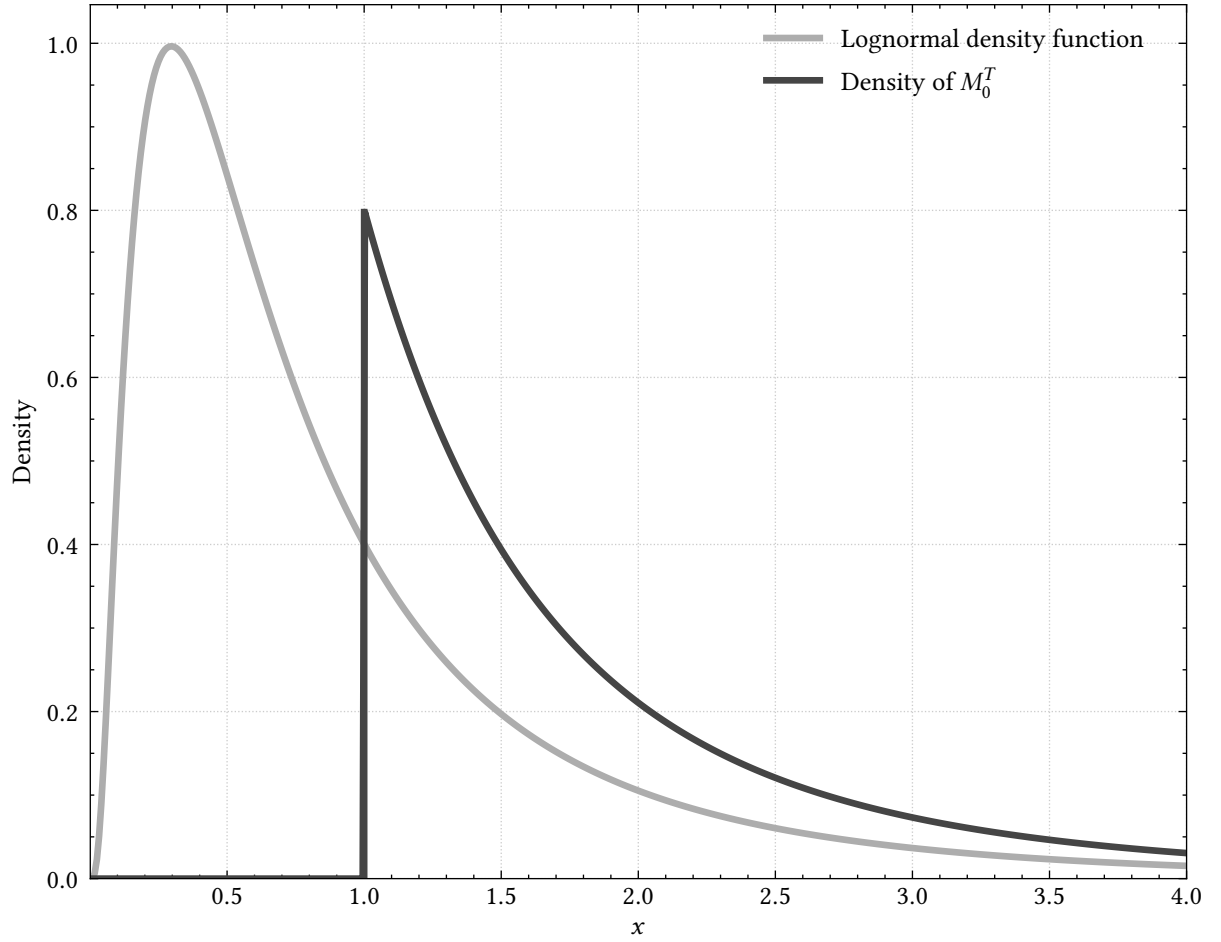


Figure 8: Density of the maximum $M_0^T = \text{Max}_{t \in [0, T]} S_t$ of geometric Brownian motion with $S_0 = 1$.

21.3 Barrier Options

The generic term "barrier options" refers to a class of options whose payoff depends on whether the underlying price hits a prespecified barrier during the option's lifetime.

21.4 Knock-out Barrier

21.4.1 Up-and-out barrier call option

Let us consider an up-and-out barrier call option with maturity T , strike price K , barrier (or call level) B , and payoff

$$C = (S_T - K)^+ \mathbb{1}_{\left\{ \max_{0 \leq t \leq T} S_t < B \right\}} = \begin{cases} S_T - K & \text{if } \max_{0 \leq t \leq T} S_t \leq B, \\ 0 & \text{if } \max_{0 \leq t \leq T} S_t > B, \end{cases}$$

with $B \geq K$.

Proposition 21.1: Up-and-Out Barrier Call Option Pricing

When $K < B$, the price

$$e^{-(T-t)r} \mathbb{1}_{\{M_0^t < B\}} \mathbb{E}^* \left[\left(\frac{S_{T-t}}{S_0} - K \right)^+ \mathbb{1}_{\left\{ x \max_{0 \leq u \leq T-t} \frac{S_u}{S_0} < B \right\}} \middle| x = S_t \right] \quad (21.6)$$

of the up-and-out barrier call option with maturity T , strike price K , and barrier level B is given by the following formula, where $\tau = T - t$:

$$\begin{aligned} C_{UO}(t) = S_t \mathbb{1}_{\{M_0^t < B\}} & \left\{ [\Phi(d_1(S_t, K)) - \Phi(d_1(S_t, B))] \right. \\ & \left. - \left(\frac{B}{S_t} \right)^{1 + \frac{2r}{\sigma^2}} \left[\Phi \left(d_1 \left(\frac{B^2}{S_t}, K \right) \right) - \Phi \left(d_1 \left(\frac{B^2}{S_t}, B \right) \right) \right] \right\} \\ & - e^{-r\tau} K \mathbb{1}_{\{M_0^t < B\}} \left\{ [\Phi(d_2(S_t, K)) - \Phi(d_2(S_t, B))] \right. \\ & \left. - \left(\frac{S_t}{B} \right)^{1 - \frac{2r}{\sigma^2}} \left[\Phi \left(d_2 \left(\frac{B^2}{S_t}, K \right) \right) - \Phi \left(d_2 \left(\frac{B^2}{S_t}, B \right) \right) \right] \right\}, \end{aligned} \quad (21.7)$$

If $K \geq B$:

$$C_{UO}(t, S_t) = 0. \quad (21.8)$$

where $d_{1,2}(S, X)$ corresponds to the definitions in (20.1) with spot price S , strike price K , and time to maturity τ . The price of the up-and-out barrier call option vanishes when $B \leq K$.

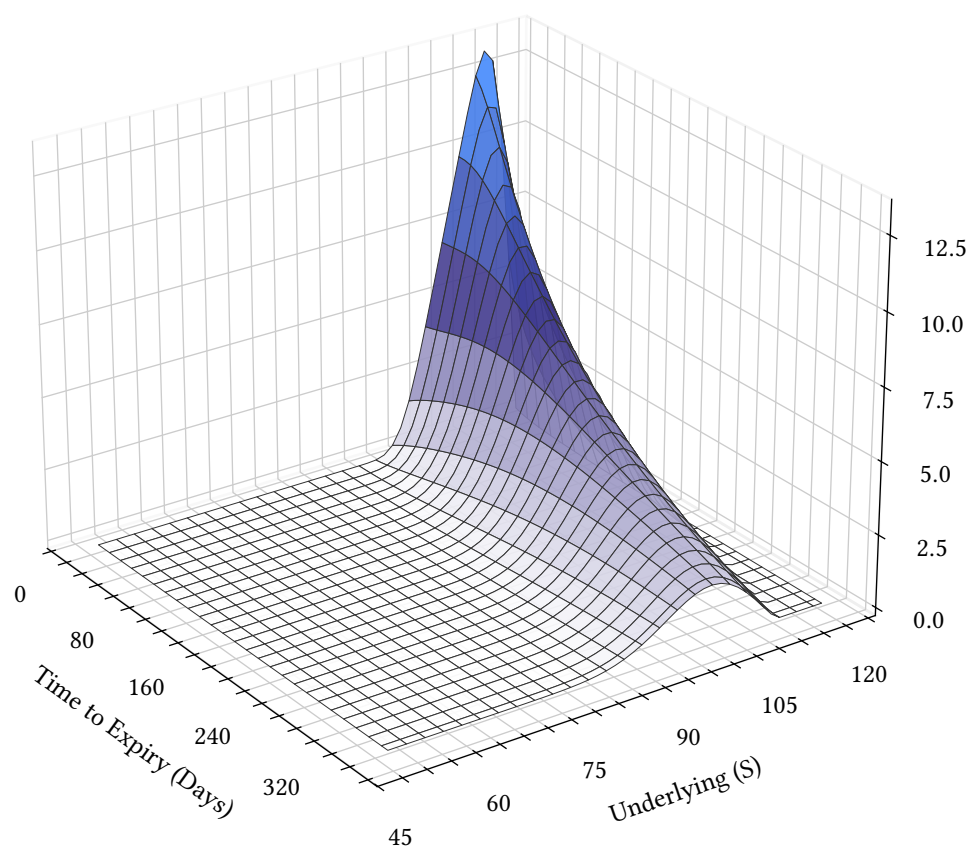


Figure 9: Graph of the up-and-out call option price

21.4.2 Up-and-out barrier put option

This option is also known as a *Callable Bear Contract*, or a Bear CBBC with no residual value, or a turbo warrant with no rebate. In this context, B denotes the call level for pricing Bear CBBCs (up-and-out barrier put options). The payoff is given by

$$P = (K - S_T)^+ \mathbb{1}_{\left\{\max_{0 \leq t \leq T} S_t < B\right\}} = \begin{cases} K - S_T & \text{if } \max_{0 \leq t \leq T} S_t \leq B, \\ 0 & \text{if } \max_{0 \leq t \leq T} S_t > B. \end{cases}$$

Proposition 21.2: Up-and-Out Barrier Put Option Pricing

The price of the up-and-out barrier put option with maturity T , strike price K , and barrier level B is given by the following formulas, where $\tau = T - t$.

If $B < K$, the price is given by

$$\begin{aligned} P_{UO}(t) = S_t \mathbb{1}_{\{M_0^t < B\}} & \left\{ -\Phi(-d_1(S_t, B)) + \left(\frac{B}{S_t}\right)^{1+\frac{2r}{\sigma^2}} \Phi\left(-d_1\left(\frac{B^2}{S_t}, B\right)\right) \right\} \\ & + Ke^{-r\tau} \mathbb{1}_{\{M_0^t < B\}} \left\{ \Phi(-d_2(S_t, B)) - \left(\frac{S_t}{B}\right)^{1-\frac{2r}{\sigma^2}} \Phi\left(-d_2\left(\frac{B^2}{S_t}, B\right)\right) \right\}. \end{aligned} \quad (21.9)$$

If $B \geq K$, the price is given by

$$\begin{aligned} P_{UO}(t) = S_t \mathbb{1}_{\{M_0^t < B\}} & \left\{ -\Phi(-d_1(S_t, K)) + \left(\frac{B}{S_t}\right)^{1+\frac{2r}{\sigma^2}} \Phi\left(-d_1\left(\frac{B^2}{S_t}, K\right)\right) \right\} \\ & + Ke^{-r\tau} \mathbb{1}_{\{M_0^t < B\}} \left\{ \Phi(-d_2(S_t, K)) - \left(\frac{S_t}{B}\right)^{1-\frac{2r}{\sigma^2}} \Phi\left(-d_2\left(\frac{B^2}{S_t}, K\right)\right) \right\}. \end{aligned} \quad (21.10)$$

This corresponds to the value of a vanilla put option minus the value of the "knock-out" component.

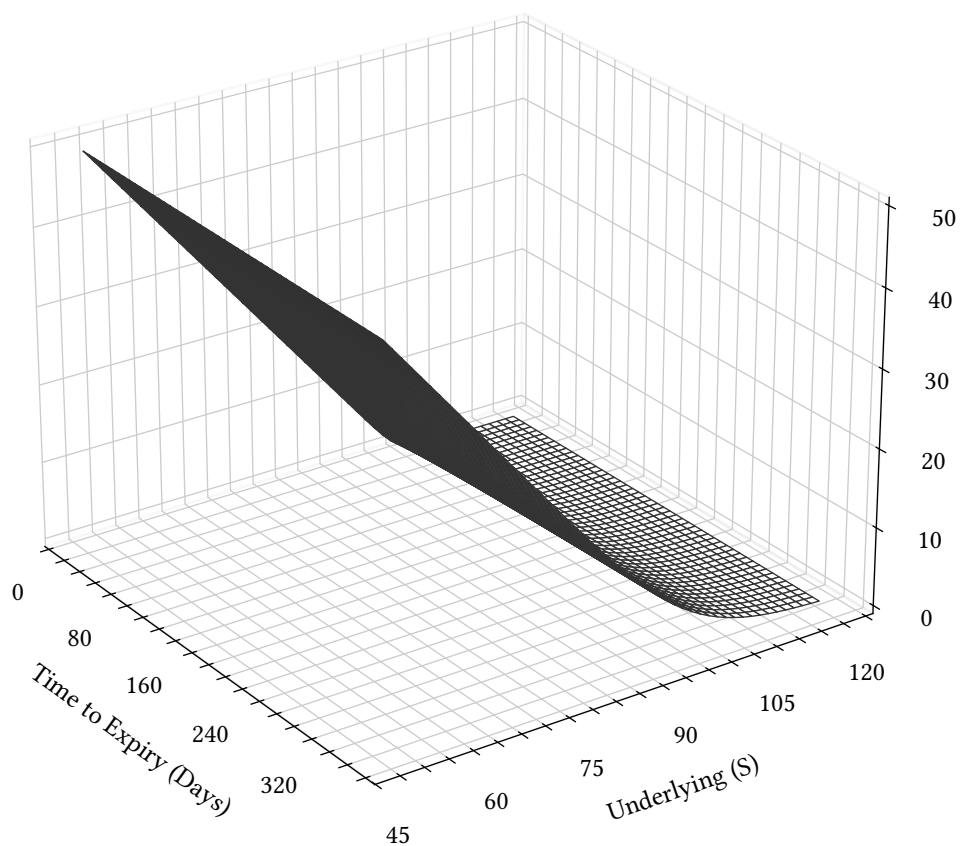


Figure 10: Graph of the up-and-out put option price

21.4.3 Down-and-out barrier call option

Let us now consider a down-and-out barrier call option on the underlying asset price S_t with exercise date T , strike price K , barrier level B , and payoff

$$C = (S_T - K)^+ \mathbb{1}_{\left\{\min_{0 \leq t \leq T} S_t > B\right\}} = \begin{cases} S_T - K & \text{if } \min_{0 \leq t \leq T} S_t > B, \\ 0 & \text{if } \min_{0 \leq t \leq T} S_t \leq B, \end{cases} \quad (21.11)$$

with $0 \leq B \leq K$. The down-and-out barrier call option is also called a *Callable Bull Contract*, or a Bull CBBC with no residual value, or a turbo warrant with no rebate, in which B denotes the call level.

Proposition 21.3: Down-and-Out Barrier Call Option Pricing

The price of the down-and-out barrier call option with maturity T , strike price K , and barrier level B is given by the following formulas, where $\tau = T - t$.

When $B \leq K$, we have:

$$\begin{aligned} C_{DO}(t, S_t) &= \mathbb{1}_{\{m_0^t > B\}} \text{Bl}(S_t, K, r, \tau, \sigma) \\ &\quad - B \mathbb{1}_{\{m_0^t > B\}} \left(\frac{B}{S_t}\right)^{1+2r/\sigma^2} \Phi\left(d_1\left(\frac{B^2}{KS_t}, \tau\right)\right) \\ &\quad + Ke^{-r\tau} \mathbb{1}_{\{m_0^t > B\}} \left(\frac{B}{S_t}\right)^{-1+2r/\sigma^2} \Phi\left(d_2\left(\frac{B^2}{KS_t}, \tau\right)\right). \end{aligned} \quad (21.12)$$

When $B \geq K$, we find:

$$\begin{aligned} C_{DO}(t, S_t) &= S_t \mathbb{1}_{\{m_0^t > B\}} \Phi\left(d_1\left(\frac{S_t}{B}, \tau\right)\right) - Ke^{-r\tau} \mathbb{1}_{\{m_0^t > B\}} \Phi\left(d_2\left(\frac{S_t}{B}, \tau\right)\right) \\ &\quad - B \mathbb{1}_{\{m_0^t > B\}} \left(\frac{B}{S_t}\right)^{2r/\sigma^2} \Phi\left(d_1\left(\frac{B}{S_t}, \tau\right)\right) \\ &\quad + Ke^{-r\tau} \mathbb{1}_{\{m_0^t > B\}} \left(\frac{S_t}{B}\right)^{1-2r/\sigma^2} \Phi\left(d_2\left(\frac{B}{S_t}, \tau\right)\right). \end{aligned} \quad (21.13)$$

Here Bl denotes the standard Black-Scholes call price, and $d_{1,2}(x, \tau)$ are defined as in ??? with the spot price term replaced by the ratio argument x . If $m_0^t \leq B$, the price is 0.

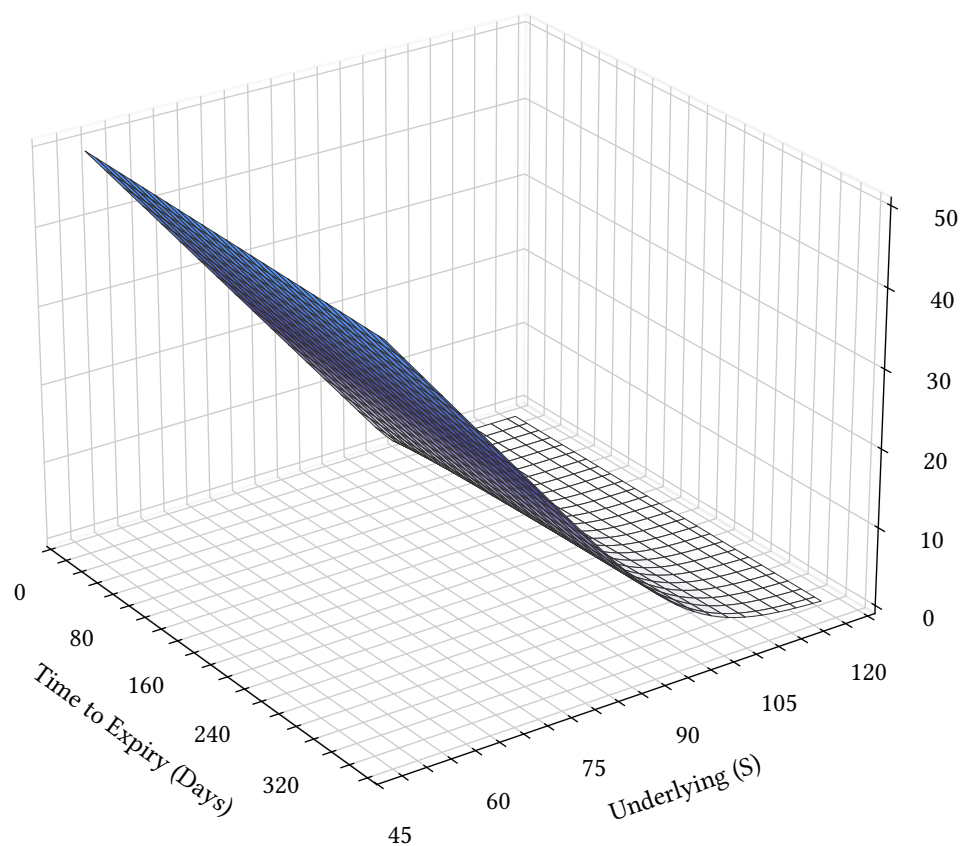


Figure 11: Graph of the down-and-out call option price

21.4.4 Down-and-out barrier put option

The down-and-out barrier put option with exercise date T , strike price K , and barrier level B has the payoff:

$$P = (K - S_T)^+ \mathbb{1}_{\{\min_{0 \leq t \leq T} S_t > B\}} = \begin{cases} K - S_T & \text{if } \min_{0 \leq t \leq T} S_t > B, \\ 0 & \text{if } \min_{0 \leq t \leq T} S_t \leq B. \end{cases} \quad (21.14)$$

For this option to have non-zero value, we typically require $S_0 > B$. Since it is a put option, the intrinsic value is positive when $S_T < K$. If $B \geq K$, the condition $S_T < K$ combined with the knock-out condition $\min S_t > B \geq K$ makes the payoff always zero (as $S_T > B \geq K \implies K - S_T < 0$). Thus, we only consider the case $B < K$.

Proposition 21.4: Down-and-Out Barrier Put Option Pricing

The price of the down-and-out barrier put option with maturity T , strike price K , and barrier level B (where $B \leq K$) is given by

$$\begin{aligned} P_{DO}(t, S_t) = & Ke^{-r\tau} \mathbb{1}_{\{m_0^t > B\}} \left[\Phi \left(-d_2 \left(\frac{S_t}{K}, \tau \right) \right) - \Phi \left(-d_2 \left(\frac{S_t}{B}, \tau \right) \right) \right] \\ & - S_t \mathbb{1}_{\{m_0^t > B\}} \left[\Phi \left(-d_1 \left(\frac{S_t}{K}, \tau \right) \right) - \Phi \left(-d_1 \left(\frac{S_t}{B}, \tau \right) \right) \right] \\ & - Ke^{-r\tau} \mathbb{1}_{\{m_0^t > B\}} \left(\frac{S_t}{B} \right)^{1-2r/\sigma^2} \left[\Phi \left(d_2 \left(\frac{B^2}{KS_t}, \tau \right) \right) - \Phi \left(d_2 \left(\frac{B}{S_t}, \tau \right) \right) \right] \\ & + S_t \mathbb{1}_{\{m_0^t > B\}} \left(\frac{B}{S_t} \right)^{1+2r/\sigma^2} \left[\Phi \left(d_1 \left(\frac{B^2}{KS_t}, \tau \right) \right) - \Phi \left(d_1 \left(\frac{B}{S_t}, \tau \right) \right) \right]. \end{aligned} \quad (21.15)$$

Here $\tau = T - t$, and $d_{1,2}(x, \tau)$ correspond to the definitions with the spot price term replaced by the ratio argument x .

If $B > K$, the price vanishes: $P_{DO}(t, S_t) = 0$.

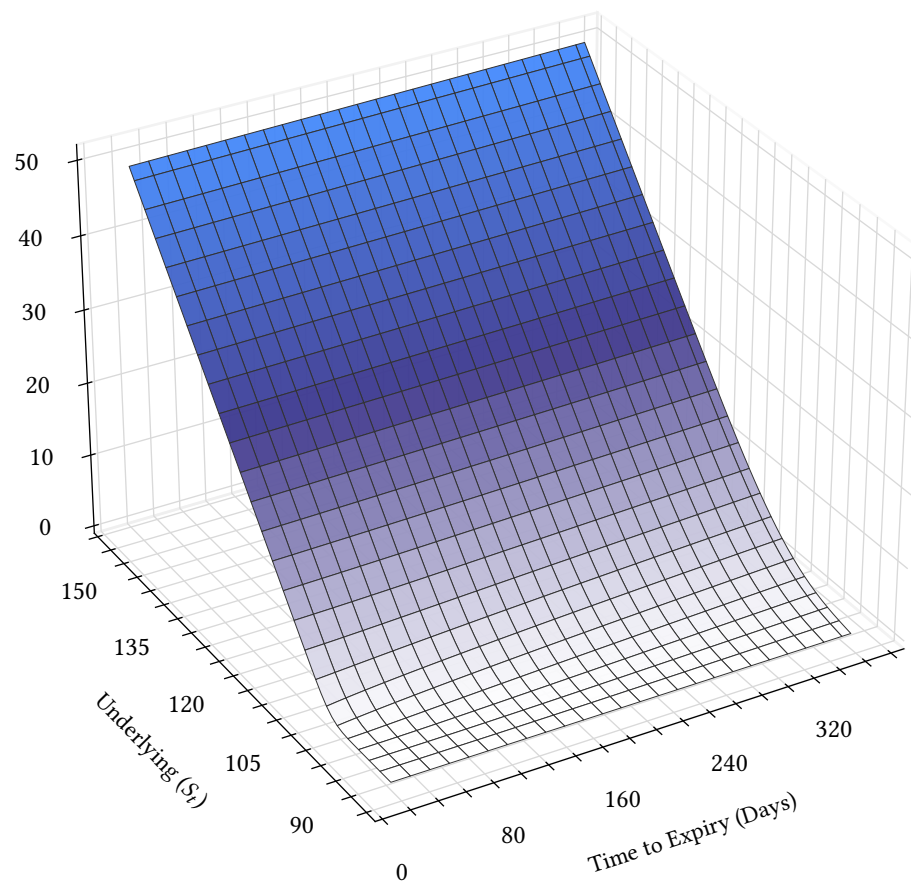


Figure 12: Graph of the down-and-out put option price

21.5 Knock-in Barrier

21.5.1 Up-and-Out Call

21.6 Lookback Options

Lookback call (resp. put) options are financial derivatives that allow their holders to exercise the option by setting the strike price at the minimum (resp. maximum) of the underlying asset price process $(S_t)_{t \in [0, T]}$ over the time interval $[0, T]$. Lookback options can be priced by PDE arguments or by computing the discounted expected values of their claim payoff C , namely $C = S_T - \min_{0 \leq t \leq T} S_t$ in the case of call options, and $C = \max_{0 \leq t \leq T} S_t - S_T$ in the case of put options.

21.7 The Lookback Put Option

The standard lookback put option gives its holder the right to sell the underlying asset at its historically highest price. In this case, the floating strike price is M_0^T and the payoff is given by the terminal value

$$C = M_0^T - S_T \quad (21.16)$$

of the drawdown process $(M_0^T - S_t)_{t \in [0, T]}$. The following pricing formula for lookback put options is a direct consequence of Proposition 10.9.

Proposition 21.5: The Price Of The Lookback Put Option

The price at time $t \in [0, T]$ of the lookback put option with payoff $M_0^T - S_T$ is given by

$$\begin{aligned}
 P_{LB}(t) &= e^{-r\tau} \mathbb{E}^* [M_0^T - S_T \mid \mathcal{F}_t] \\
 &= M_0^t e^{-r\tau} \Phi(-d_2(S_t, M_0^t)) \\
 &\quad + S_t \left(1 + \frac{\sigma^2}{2r}\right) \Phi(d_1(S_t, M_0^t)) \\
 &\quad - S_t e^{-r\tau} \frac{\sigma^2}{2r} \left(\frac{M_0^t}{S_t}\right)^{2r/\sigma^2} \Phi(-d_2(M_0^t, S_t)) - S_t,
 \end{aligned} \tag{21.17}$$

where $\tau = T - t$, and $d_{1,2}(\cdot, \cdot)$ correspond to the definitions in (21.1) with the fixed strike K replaced by the running maximum M_0^t (or S_t where arguments are inverted).

21.8 Floating Strike Lookback Option

We begin with a geometric brownian motion asset price, which may be written as in

21.8.1 Floating Strike Lookback Option

21.9 Asian Options

An *Asian option* (or *average option*) defines a class of options, whether European or American style, whose terminal payoff depends on the average asset value over a specific period within the option's life. Due to this averaging feature, Asian options are particularly suitable for thinly traded assets or commodities. Unlike standard options, they are typically less expensive and offer greater robustness against price manipulation near the expiry date.

Asian options were first traded in Tokyo in 1987, and have become particularly popular in commodities trading.

Let T denote the exercise date, and let $0 \leq T_0 < T$ represent the commencement of the averaging period. The payoff at expiry for an Asian call option is defined as

$$C_T^A := (A_S(T_0, T) - K)^+, \tag{21.18}$$

where

$$A_S(T_0, T) = \frac{1}{T - T_0} \int_{T_0}^T S_u \, du \quad (21.19)$$

represents the arithmetic average of the asset price over the time interval $[T_0, T]$. Here, K denotes the fixed strike price, and the asset price S is assumed to follow a geometric Brownian motion.

The main difficulty in pricing and hedging Asian options is due to the fact that the random variable $A(T_0, T)$ does not have a lognormal distribution. This feature makes the task of finding an explicit formula for the price of an Asian option surprisingly involved.

For this reason, early studies of Asian options were based either on approximations or on the direct application of the Monte Carlo method. The numerical approach to the valuation of Asian options, is based on the approximation of the arithmetic average using the geometric average. Note first that it is natural to substitute for the continuous-time average $A(T_0, T)$ its discrete-time counterpart

$$A_S^n(T_0, T) = \frac{1}{n} \sum_{i=0}^{n-1} S_{T_i}, \quad (21.20)$$

where $T_i = T_0 + i(T - T_0)/n$. Furthermore, the arithmetic average $A_S^n(T_0, T)$ can be replaced with the geometric average, denoted by $G_S^n(T_0, T)$, in what follows. Recall that the random variables $S_{T_i}, i = 1, \dots, n$ are given explicitly by the expression

$$S_{T_i} = S_{T_0} \exp \left(\sigma(W_{T_i}^* - W_{T_0}^*) + \left(r - \frac{1}{2}\sigma^2 \right) (T_i - T_0) \right), \quad (21.21)$$

where W^* is a standard Brownian motion under the martingale measure $\mathbb{P}^*$. Hence the geometric average admits the following representation

$$G_S^n(T_0, T) = \left(\prod_{i=0}^{n-1} S_{T_i} \right)^{1/n} = c S_{T_0} \exp \left(\frac{\sigma}{n} \sum_{i=0}^{n-1} (n - i - 1) (W_{T_{i+1}}^* - W_{T_i}^*) \right) \quad (21.22)$$

for a strictly positive constant c . In view of the independence of increments of the Brownian motion, the last formula makes clear that the geometric average $G_S^n(T_0, T)$ has a lognormal distribution under $\mathbb{P}^*$. The approximate Black-Scholes-like formula for the price of an Asian

call option can thus be easily found by the direct evaluation of the conditional expectation

$$\tilde{C}_{T_0}^n = e^{-r(T-T_0)} \mathbb{E}_{\mathbb{P}^*} \left((G_S^n(T_0, T) - K)^+ \mid \mathcal{F}_{T_0} \right). \quad (21.23)$$

It appears, however, that such an approach significantly underprices Asian call options. To overcome this deficiency, one may directly approximate the true distribution of the arithmetic average using an approximate distribution, typically a lognormal distribution with the appropriate parameters

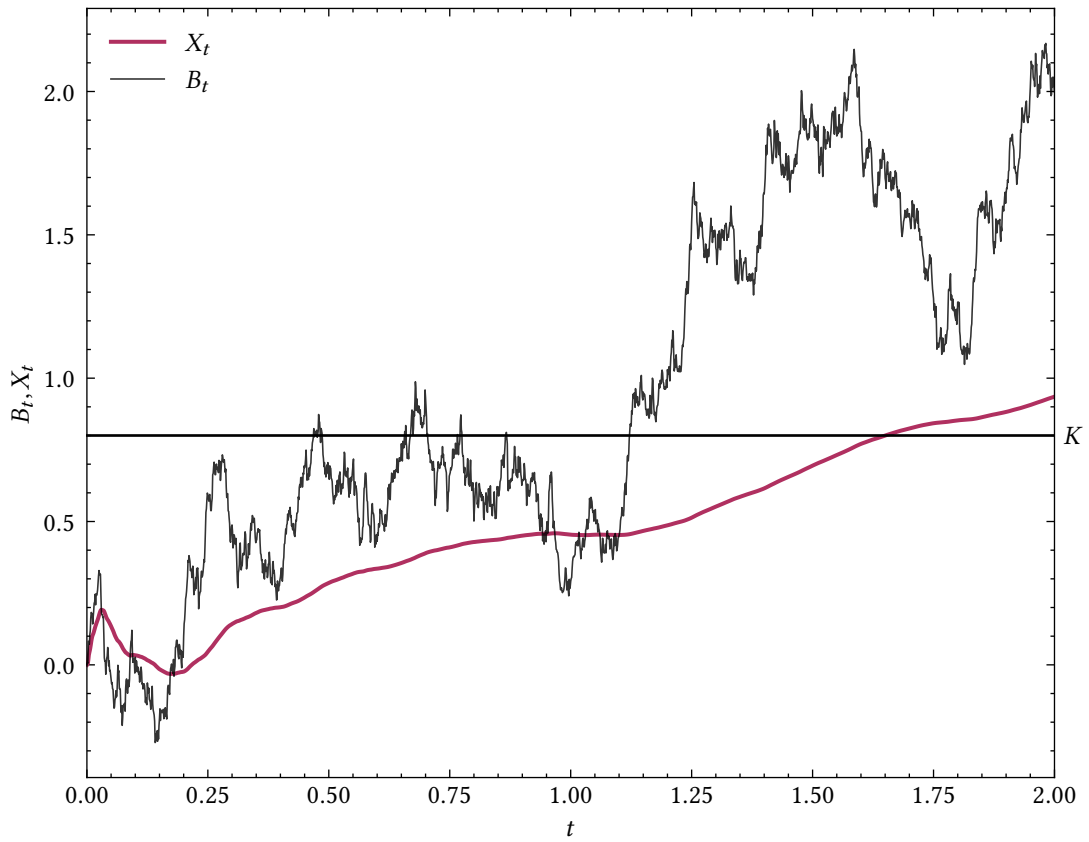


Figure 13: Brownian motion B_t and its moving average X_t

22 American Options

23 Basket Options

Appendices

A Existence of Lebesgue Measure

B Linear Algebra and Functional Analysis

C Multivariate Differentiation

D Probability and Statistics

E Python Primer